

高等数学

概念

极限 = 例 1.1, 2

$$\lim_{x,y \rightarrow 0} \frac{xy}{x+y} \text{ 不存在}$$

$$\lim_{x,y \rightarrow 0} \frac{xy}{x^2+y^2} \text{ 不存在}$$

$$\lim_{x,y \rightarrow 0} \frac{xy}{\sqrt{x^2+y^2}} = 0$$

常用极限存在/不存在

连续 例 3.4

偏导数 例 5.6

偏导数存在
偏导数连续 $\Rightarrow$ 可微 $\Rightarrow$ 连续 $\Rightarrow$ 极限存在(全方向)
 $\Downarrow$
方向导数存在(某方向单侧)

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

写全增量

$$A\Delta x + B\Delta y$$

写线性增量

$$\lim_{\Delta x, \Delta y \rightarrow 0} \frac{\Delta z - (A\Delta x + B\Delta y)}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}$$

做极限

验证是否可微

例 3.4, 7

可微

5'

已知可微, 用 $\Delta z = A\Delta x + B\Delta y + o(\rho)$

偏导数连续

5'

1. 定义法求偏导数
2. 公式法求偏导数
3. 公式法结果取极限与定义法结果比较

概要

偏导数存在(某方向双侧)
偏导数连续 $\Rightarrow$ 可微 $\Rightarrow$ 连续 $\Rightarrow$ 极限存在(全方向)
方向导数存在(某方向单侧) (双数第一, 是第17讲)

注意

$$f'_x(x, y_0)$$

eg. $f(x, 1) = 0$
 $f'_x(x, 1) = ?$
 $= (f(x, 1))'_x = 0$

可以先代值再求偏导!
例 14.23

$$f'_y(0, y) = (f(0, y))'_y \quad \text{例 13.30}$$

eg. 已知 $f'_x(1, y) = 2y$ 例 13.6
则 $f''_{xy}(1, y) = ?$ 2

计算

复合函数求导

5' 12'

链式求导规则 例 8, 9, 10, 39

全导数 例 12

全微分形式不变性 例 13.14

隐函数存在定理

$$F(x, y, z) = 0$$

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} \quad \text{例 11, 13, 14, 15, 16}$$

一个方程的情形

隐函数求导

5' 12'

$$F(x, y, z) = 0 \quad \begin{cases} y = y(x) \\ z = z(x) \end{cases}$$

$$G(x, y, z) = 0$$

$$x, y, z \rightarrow x$$

$$\frac{\partial u}{\partial x} = -\frac{\frac{\partial(F, G)}{\partial(z, u)}}{\frac{\partial(F, G)}{\partial(z, u)}}$$

$$\frac{\partial v}{\partial x} = -\frac{\frac{\partial(F, G)}{\partial(y, u)}}{\frac{\partial(F, G)}{\partial(y, u)}}$$

方程组的情形 例 12

$$\begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix} = \frac{\partial(F, G)}{\partial(u, v)}$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + R$$

$$f(x,y) = f(x_0,y_0) + [f'_x, f'_y]_{x_0} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \frac{1}{2!} [\Delta x \ \Delta y] \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{yx} & f''_{yy} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + R$$

多元函数的泰勒公式 $f(x,y) = f(x_0,y_0) + [f'_x, f'_y]_{x_0} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} + \frac{1}{2!} [f''_{xx}(x-x_0)^2 + f''_{yy}(y-y_0)^2 + 2f''_{xy}(x-x_0)(y-y_0)]$ 例 17

多元函数的极最值

12'

例 21

无条件极值

$$A > 0 \quad \begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} > 0$$

$$AC - B^2 > 0 \quad \begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} > 0$$

正定 (极小值)

$$A < 0 \quad \begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} < 0$$

$$AC - B^2 > 0 \quad \begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} < 0$$

负定 (极大值)

$$\begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} < 0$$

非极值

$$\begin{vmatrix} f''_{xx} & f''_{yx} \\ f''_{yx} & f''_{yy} \end{vmatrix}_{X_0} = 0$$

失效

例 22~25

条件极值与拉氏成数法

目标函数 $f(x,y,z)$

$$\text{约束条件} \quad \begin{cases} \phi_1(x,y,z) = 0 \\ \phi_2(x,y,z) = 0 \end{cases}$$

$$\text{辅助函数} \quad F(x,y,z,\lambda,\mu) = F(x,y,z) + \lambda\phi_1(x,y,z) + \mu\phi_2(x,y,z)$$

$$\text{求偏导} \quad F'_x = F'_y = F'_z = F'_\lambda = F'_\mu = 0$$

得到极值 (一般求最值)

- 求解方法
- 消元带入
 - 观察法
 - 字母的轮换对称性 (注意是 $x=y$ 还是 $x=-y$)
 - 带根号, 用里边的
 - 带绝对值, 用平方

已知偏导数, 求 $z=f(x,y)$

例 27, 28

给出变换, 化已知偏微分方程为常微分方程, 求 $f(u)$

例 29, 30

给出变换, 化已知偏微分方程为指定偏微分方程并求其反问题

例 31

偏微分方程

12'

应用

例 13.3



函数 $f(x, y) = \begin{cases} \frac{\sqrt[3]{1+xy}-1}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$ 在点 $(0, 0)$ 处().

- (A) 不连续 (B) 连续但偏导数不存在
(C) 连续, 偏导数存在但不可微 (D) 可微



$$\textcircled{1} 0 < \lim_{x, y \rightarrow 0} f(x, y) = \lim_{x, y \rightarrow 0} \frac{\sqrt[3]{1+xy}-1}{\sqrt{x^2+y^2}} = \lim_{x, y \rightarrow 0} \frac{\frac{1}{3}xy}{\sqrt{x^2+y^2}} \leq \lim_{x, y \rightarrow 0} \frac{1}{3\sqrt{2}} \sqrt{xy} = 0$$

$$\lim_{x, y \rightarrow 0} f(x, y) = 0 = f(0, 0) \quad \sqrt{x^2+y^2} \leq \sqrt{2xy}$$

$$\textcircled{2} f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x \rightarrow 0} \frac{0}{x} = 0 \quad \text{同理 } f'_y(0, 0) = 0$$

$$\textcircled{3} \text{可微: } \lim_{x, y \rightarrow 0} \frac{f(x, y) - f(0, 0) - [f'_x(0, 0)(x-0) + f'_y(0, 0)(y-0)]}{\sqrt{x^2+y^2}} = \lim_{x, y \rightarrow 0} \frac{\sqrt[3]{1+xy}-1}{\sqrt{x^2+y^2}} = \lim_{x, y \rightarrow 0} \frac{1}{3} \frac{xy}{\sqrt{x^2+y^2}} = \lim_{x \rightarrow 0} \frac{k}{3} \frac{1}{1+k^2} \text{ 不可微 (C)}$$

例 13.1

例 13.1 证明: 极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{\sqrt{x+y+1}-1}$ 不存在.

$$\text{证: } \lim_{x, y \rightarrow 0} \frac{2xy}{x+y} \quad y=kx \quad \lim_{x \rightarrow 0} \frac{2kx^2}{(1+k)x} = 0 \quad y=kx^2-x \quad \lim_{x \rightarrow 0} \frac{2kx^3-x^2}{kx^2} = -\frac{1}{k}$$

例 13.2

例 13.2 求极限 $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \left(\frac{xy}{x^2+y^2} \right)^{x^2}$.

$$\lim_{x, y \rightarrow \infty} \left(\frac{xy}{x^2+y^2} \right)^{x^2} = \lim_{x, y \rightarrow \infty} e^{x^2 \ln \left(\frac{xy}{x^2+y^2} \right)} \leq \lim_{x, y \rightarrow \infty} e^{x^2 \ln \frac{xy}{2xy}} = \lim_{x, y \rightarrow \infty} e^{-\ln 2 x^2} = 0$$

例 13.4

例 13.4 设连续函数 $z = f(x, y)$ 满足 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} = 0$.

则 $dz|_{(0,1)} = \underline{\hspace{2cm}}$.

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}}$$

$$\lim_{\substack{x \rightarrow 0 \\ y=1}} \frac{f(x, 1) - 2x - 1}{x - 0} = 0 \quad f'_x(x, 1) = 2$$

$$\frac{f(x, 1) - 1}{x - 0} = 2$$

$$\lim_{\substack{x=0 \\ y \rightarrow 1}} \frac{f(0, y) + y - 2}{y - 1} = 0$$

$$\frac{f(0, y) - 1}{y - 1} = -1$$

$$f'_y(0, y) = -1$$

$$dz = 2dx - dy$$

例 13.5

例 13.5 已知 $f(x, y) = e^{\sqrt{x^2+y^4}}$, 则().

- (A) $f'_x(0,0), f'_y(0,0)$ 都存在 (B) $f'_x(0,0)$ 不存在, $f'_y(0,0)$ 存在
(C) $f'_x(0,0)$ 存在, $f'_y(0,0)$ 不存在 (D) $f'_x(0,0), f'_y(0,0)$ 都不存在

$$f(x, y) = e^{\sqrt{x^2+y^4}}$$

$$f(x, 0) = e^{x^2} \quad f(x, 0) = e^{x^2} \Rightarrow \text{A, B}$$

例 13.6

例 13.6 设函数 $f(x, y) = \int_0^{xy} e^{t^2} dt$, 则 $\left. \frac{\partial^2 f}{\partial x \partial y} \right|_{(1,1)} =$.

$$\left[\int_0^{xy} e^{t^2} dt \right]_x = \left[e^{x^2 y^2} y \right]$$

$$(e^{x^2 y^2})' = e^{x^2 y^2} + e^{x^2 y^2} \cdot 2y = 3e^{x^2 y^2}$$

$$\left[\int_0^{xy} e^{t^2} dt \right]_x = \left[\int_0^{x^2 y^2} e^{u^2} \frac{du}{2y} \right]_x = \left[\frac{1}{2y} \int_0^{x^2 y^2} e^{u^2} du \right]_x = -\frac{1}{2} x^{-\frac{3}{2}} \int_0^{x^2 y^2} e^{u^2} du + \frac{1}{2y} e^{x^2 y^2} \cdot \frac{3}{2} x^{\frac{1}{2}}$$

$$\left(-\frac{1}{2} x^{-\frac{3}{2}} \int_0^{x^2 y^2} e^{u^2} du + \frac{3y}{2x} e^{x^2 y^2} \right) \Big|_{x=1} = \left(-\frac{1}{2} \int_0^y e^{u^2} du + \frac{3}{2} e^{y^2} \right) \Big|_{y=1} = -\frac{1}{2} e^{y^2} + 3e^{y^2} + \frac{3}{2} e^{y^2} \Big|_{y=1} = -\frac{1}{2} e + 3e + \frac{3}{2} e = 4e$$

例 13.7

例 13.7 二元函数 $f(x, y)$ 在点 $(0, 0)$ 处可微的一个充分条件是(C).

- (A) $\lim_{(x,y) \rightarrow (0,0)} [f(x, y) - f(0, 0)] = 0$
(B) $\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0$, 且 $\lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = 0$
(C) $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0)}{\sqrt{x^2 + y^2}} = 0$
(D) $\lim_{x \rightarrow 0} [f'_x(x, 0) - f'_x(0, 0)] = 0$, 且 $\lim_{y \rightarrow 0} [f'_y(0, y) - f'_y(0, 0)] = 0$

例 13.8

例 13.8 设 $z = (x^2 + y^2)^{xy}$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

$$u = x^2 + y^2 \quad z = u^v = e^{v \ln u}$$

$$v = xy$$

$$\begin{cases} \frac{\partial u}{\partial x} = 2x & \frac{\partial u}{\partial y} = 2y & \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = v u^{v-1} \cdot 2x + u^v \cdot \ln u \cdot y \\ \frac{\partial z}{\partial y} = y & \frac{\partial v}{\partial y} = x & \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = v u^{v-1} \cdot 2y + u^v \cdot \ln u \cdot x \end{cases}$$

例 13.9

例 13.9 设 $z = f(x^2 - y^2, \cos(xy))$, 其中 f 具有一阶连续偏导数, $x = r \cos \theta = 0$,

$y = r \sin \theta = 0$. 求 $\frac{\partial z}{\partial r}, \frac{\partial z}{\partial \theta}$.

$$z = f(x^2 - y^2, \cos(xy))$$

$$\begin{cases} x = r \cos \theta & x'_r = \cos \theta & x'_\theta = -r \sin \theta \\ y = r \sin \theta & y'_r = \sin \theta & y'_\theta = r \cos \theta \end{cases}$$

$$\frac{\partial z}{\partial r} = f'_1 \cdot (2x \cdot \cos \theta - 2y \cdot \sin \theta) + f'_2 \cdot (-y \sin(xy) \cdot \cos \theta - x \sin(xy) \cdot \sin \theta)$$

$$\frac{\partial z}{\partial \theta} = f_1' (2x \cdot (-\sin \theta) - 2y (\sin \theta)) + f_2' (-y \sin(xy) \cdot (-\cos \theta) - x \sin(xy) \cdot (\sin \theta))$$

例 13.10

例 13.10 设 $z = f(x^2 - y^2, e^{xy})$, 其中 f 具有连续二阶偏导数, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x \partial y}$.

$$\frac{\partial z}{\partial x} = f_1' \cdot 2x + f_2' \cdot e^{xy} \cdot y$$

$$\frac{\partial z}{\partial y} = f_1' \cdot (-2y) + f_2' \cdot e^{xy} \cdot x$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2x \cdot [f_1''(-2y) + f_1'' \cdot e^{xy} \cdot x] + e^{xy} \cdot y \cdot [f_2''(-2y) + f_2'' \cdot e^{xy} \cdot x] + f_2' [e^{xy} + e^{xy} \cdot xy]$$

例 13.9

例 13.9 已知函数 $f(u, v)$ 具有二阶连续偏导数, $f(1, 1) = 2$ 是 $f(u, v)$ 的极值, $z =$

$$f[x+y, f(x, y)], \text{ 求 } \left. \frac{\partial^2 z}{\partial x \partial y} \right|_{(1,1)}.$$

$$z = f[x+y, f(x, y)]$$

$$\frac{\partial z}{\partial x} = f_1'[x+y, f(x, y)] + f_2'[x+y, f(x, y)] \cdot f_1'(x, y)$$

$$\frac{\partial^2 z}{\partial x \partial y} = f_{11}''[x+y, f(x, y)] + f_{12}''[x+y, f(x, y)] \cdot f_1'(x, y) + f_{21}''[x+y, f(x, y)] \cdot f_2'(x, y) + f_{22}''[x+y, f(x, y)] \cdot f_1'(x, y) \cdot f_2'(x, y)$$

$$\left. \frac{\partial^2 z}{\partial x \partial y} \right|_{(1,1)} = f_{11}''(2, 2) + f_{12}''(2, 2) f_2''(1, 1)$$

例 13.16

例 13.16 设函数 $z = z(x, y)$ 是由方程 $F(x + zy^{-1}, y + zx^{-1}) = 0$ 所确定, 其中 F 具有连续偏导数, 且 $xF_1' + yF_2' \neq 0$, 证明

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy.$$

$$F(x + zy^{-1}, y + zx^{-1}) = 0$$

$$\frac{\partial z}{\partial x} = - \frac{F_1'}{F_2'} = - \frac{F_1' + F_2' \cdot \frac{-z}{x^2}}{F_1' \cdot \frac{1}{y} + F_2' \cdot \frac{1}{x}}$$

$$\frac{\partial z}{\partial y} = - \frac{F_2'}{F_1'} = - \frac{F_1' \cdot \frac{-z}{y^2} + F_2'}{F_1' \cdot \frac{1}{y} + F_2' \cdot \frac{1}{x}}$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = - \frac{xF_1' - F_2' \frac{z}{x}}{F_1' \frac{1}{y} + F_2' \frac{1}{x}} - \frac{F_1' \frac{-z}{y} + yF_2'}{\frac{1}{y} F_1' + F_2' \frac{1}{x}} = \frac{xF_1' + yF_2' - z(\frac{F_1'}{y} - \frac{F_2'}{x})}{-(\frac{F_1'}{y} + \frac{F_2'}{x})} = z - xy$$

例 13.14

例 13.14 若函数 $z = z(x, y)$ 由方程 $e^{x+2y+3z} + xyz = 1$ 确定, 则 $dz|_{(0,0)} =$ _____.

$$e^{x+2y+3z} + xyz = 1$$

$$x=0, y=0 \Rightarrow$$

$$e^{3z} = 1$$

方法一. 微分法:

$$e^{x+2y+3z} \cdot (1 + 3 \frac{\partial z}{\partial x}) + yz + xy \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{1}{3}$$

$$e^{x+2y+3z} \cdot (2 + 3 \frac{\partial z}{\partial y}) + xz + xy \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \frac{\partial z}{\partial y} = -\frac{2}{3}$$

$$dz|_{(0,0)} = -\frac{1}{3}dx - \frac{2}{3}dy$$

方法二. 公式法

$$F(x, y, z) = e^{x+2y+3z} + xyz - 1 = 0$$

$$\frac{\partial z}{\partial x} = - \frac{F_1'}{F_2'} = - \frac{e^{x+2y+3z} + yz}{3e^{x+2y+3z} + xy} (-1) |_{(0,0)} = -\frac{1}{3}$$

$$\frac{\partial z}{\partial y} = - \frac{F_2'}{F_1'} = - \frac{2e^{x+2y+3z} + xz}{3e^{x+2y+3z} + xy} (-1) |_{(0,0)} = -\frac{2}{3}$$

方法三 全微分形式不变性

$$d(e^{x+2y+3z} + xy^2z) = 0$$

$$\rightarrow e^{x+2y+3z} d(x+2y+3z) + d(xy^2z) = 0$$

$$e^{x+2y+3z} (1 \cdot dx + 2dy + 3dz) + y^2z dx + x \cdot 2y \cdot dz + xy^2 dz = 0$$

(0,0)处 $dx + 2dy + 3dz = 0$

$$dz = -\frac{1}{3}dx - \frac{2}{3}dy$$

例 13.15

例 13.15 设 $F(x, y, z)$ 连续可微, $F'_x \cdot F'_y \cdot F'_z \neq 0$, 方程 $F(x, y, z) = 0$ 可确定连续可微的隐函数 $z = z(x, y)$, $y = y(z, x)$, $x = x(y, z)$, 则().

- (A) $\frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial z} = -3$ (B) $\frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial z} = 3$
(C) $\frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial z} = -1$ (D) $\frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial z} = 1$

$$\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} \quad (C)$$

例 13.12

例 13.12 设 $y = y(x)$, $z = z(x)$ 是由方程 $z = xf(x+y)$ 和 $F(x, y, z) = 0$ 所确定的函数, 其中 f 和 F 分别具有一阶连续导数和一阶连续偏导数, 且 $F'_y + xf'F'_z \neq 0$, 求 $\frac{dz}{dx}$.

$$\begin{aligned} & \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = z - xf(x+y) = 0 \end{cases} \\ & \frac{dz}{dx} = -\frac{\frac{\partial(F, G)}{\partial(y, x)}}{\frac{\partial(F, G)}{\partial(y, z)}} = -\frac{\begin{vmatrix} \frac{\partial F}{\partial y} & \frac{\partial F}{\partial x} \\ \frac{\partial G}{\partial y} & \frac{\partial G}{\partial x} \end{vmatrix}}{\begin{vmatrix} \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \end{vmatrix}} = -\frac{F'_2 \cdot (-f(x+y)) - F'_1(-xf'(x+y))}{F'_2 - F'_3(-xf'(x+y))} \end{aligned}$$

例 13.11

例 13.11 设方程 $x + y^2 + \sin(xy) = 0$, 则在点 $(0, 0)$ 的某邻域内, 该方程(B).

- (A) 只可以确定一个具有连续导数的隐函数 $y = y(x)$
(B) 只可以确定一个具有连续导数的隐函数 $x = x(y)$
(C) 可以确定两个具有连续导数的隐函数 $x = x(y)$ 和 $y = y(x)$
(D) 不可以确定任何一个具有连续导数的隐函数

$$F(x, y) = x + y^2 + \sin(xy) = 0$$

$$\begin{cases} F'_x = 1 + y \cos xy \stackrel{(0,0)}{=} 1 \\ F'_y = 2y + x \cos xy \stackrel{(0,0)}{=} 0 \end{cases} \quad -\frac{F'_x}{F'_y} = y'_x \text{ 不存在}$$

例 13.13

例 13.13 设 $x \cos y + y \cos z + z \cos x = 1$, 且 $y \sin z - \cos x \neq 0$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

$$\begin{aligned} & F(x, y, z) = x \cos y + y \cos z + z \cos x - 1 = 0 \\ & \begin{cases} F'_x = \cos y - z \sin x & \frac{\partial z}{\partial x} = -\frac{\cos y - z \sin x}{-y \sin z + \cos x} \\ F'_y = -x \sin y + \cos z & \frac{\partial z}{\partial y} = -\frac{-x \sin y + \cos z}{-y \sin z + \cos x} \\ F'_z = -y \sin z + \cos x \end{cases} \end{aligned}$$

$$\begin{aligned} f(e,0) &= 1 \\ f'_x &= yx^{y-1} \Big|_{(e,0)} = 0 \\ f'_y &= x^y \ln x \Big|_{(e,0)} = 1 \\ f''_{xx} &= y(y-1)x^{y-2} \Big|_{(e,0)} = 0 \\ f''_{yy} &= x^y \ln^2 x \Big|_{(e,0)} = 1 \\ f''_{xy} &= x^y + yx^{y-1} \ln x \Big|_{(e,0)} = e^{-1} \end{aligned}$$

例 13.17

例 13.17 (仅数学一) 二元函数 $f(x,y) = x^y$ 在点 $(e,0)$ 处的二阶 (即 $n=2$) 泰勒展开

式 (不要求写余项) 为 _____.

$$\begin{aligned} f(x,y) &= f(x_0,y_0) + [f'_x, f'_y]_{x_0} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \frac{1}{2} [\Delta x \ \Delta y] \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{xy} & f''_{yy} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \\ f(x,y) &= f(e,0) + f'_x(e,0)(x-e) + f'_y(e,0)(y-0) \\ &\quad + \frac{1}{2} \left[f''_{xx}(e,0)(x-e)^2 + 2f''_{xy}(e,0)(x-e)(y-0) + f''_{yy}(e,0)(y-0)^2 \right] \\ &= 1 + y + \frac{1}{2} [2e^1(x-e)y + y^2] \end{aligned}$$

例 13.18

例 13.18 已知函数 $f(x,y)$ 在点 $(0,0)$ 的某邻域内连续, 且

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x,y) - axy}{(x^2+y^2)^2} = 1,$$

其中 a 为非零常数, 则 $f(0,0)$ ().

(A) 是极大值

(B) 是极小值

(C) 不是极值

(D) 是否取极值与 a 有关

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x,y) - axy}{(x^2+y^2)^2} = 1$$

$$f(x,y) - axy = (1 + \alpha)(x^2+y^2)^2$$

$$f(x,y) = axy + (1 + \alpha)(x^2+y^2)^2$$

$$f(x,y) = axy + (x^2+y^2)^2 + \alpha(x^2+y^2)^2$$

$$f'_x = ay + 4x(x^2+y^2) + 4\alpha x(x^2+y^2)$$

$$f'_y = ax + 4y(x^2+y^2) + 4\alpha y(x^2+y^2)$$

$$f''_{xx} = 4(x^2+y^2) + 8x^2 + 4\alpha(x^2+y^2) + 8\alpha x^2 > 0$$

$$f''_{yy} = 4(x^2+y^2) + 8y^2 + 4\alpha(x^2+y^2) + 8\alpha y^2$$

$$f''_{xy} = a + 8xy + 4\alpha + 8\alpha xy$$

$$AC - B^2 = [4\alpha(x^2+y^2) + 8x^2][4\alpha(x^2+y^2) + 8y^2] - (8xy + a)^2$$

$$= (12x^2 + 4y^2)(12y^2 + 4x^2)$$

$\Rightarrow$ 在 $y=x$ 上

$$f(x,x) = ax^2 + 4x^4 + \alpha \cdot 4x^4$$

$$x \rightarrow 0 \text{ 时 } f(x,x) \sim ax^2 \quad f(x,x) \text{ 与 } a \text{ 同号}$$

在 $y=-x$ 上

$$f(x,-x) = -ax^2 + 4x^4 + \alpha \cdot 4x^4$$

$$x \rightarrow 0 \text{ 时 } f(x,-x) \sim -ax^2 \quad f(x,-x) \text{ 与 } a \text{ 异号}$$

C

例 13.19

例 13.19 设 $f(x)$ 为二阶可导函数, 且 $x=0$ 是 $f(x)$ 的驻点, 则二元函数 $z = f(x)f(y)$

在点 $(0,0)$ 处取得极大值的一个充分条件是 ().

(A) $f(0) < 0, f''(0) > 0$

(B) $f(0) < 0, f''(0) < 0$

(C) $f(0) > 0, f''(0) > 0$

(D) $f(0) = 0, f''(0) \neq 0$

$$z'_x = f(y) \cdot f'(x)$$

$$z''_{xx} = f(y) \cdot f''(x) = A < 0$$

$$AC - B^2 = AC > 0$$

$$z'_y = f'(y) \cdot f(x)$$

$$z''_{yy} = f''(y) \cdot f(x) = C$$

$$z''_{xy} = f'(y) \cdot f'(x) = B = 0$$

$$\Rightarrow \begin{cases} A < 0 \\ C < 0 \end{cases}$$

例 13.20

例 13.20 证明: 函数 $z = (1+e^y)\cos x - ye^y$ 有无穷多个极大值点而无极小值点.

$$z'_x = -\sin x (1+e^y) = 0 \Rightarrow \sin x = 0$$

$$z'_y = e^y \cos x - e^y - ye^y = 0 \Rightarrow y = \cos x - 1$$

$$z''_{xx} = -\cos x (1+e^y) = A$$

$$z''_{yy} = e^y (\cos x - 2 - y) = C$$

$$z''_{xy} = -\sin x \cdot e^y = B$$

$$AC - B^2 = -\cos x (1+e^y) e^y (\cos x - 2 - y) - \sin^2 x \cdot e^{2y}$$

$$= \cos x (1+e^y) e^y = (y+1)(1+e^y) e^y > 0$$

$$\Rightarrow y=0$$

$$A = -(y+1)(1+e^y) = -2 < 0$$

$$\Rightarrow y=0, \sin x=0 \text{ 时 极大}$$

例 13.21

例 13.21 求函数 $f(x, y) = 2\ln|x| + \frac{(x-1)^2 + y^2}{2x^2}$ 的极值.

$$\begin{cases} f'_x = \frac{2}{x} + \frac{2(x-1) \cdot 2x^2 - [(x-1)^2 + y^2] \cdot 2x}{4x^4} \\ = \frac{2x^2 + x - 1 - y^2}{x^3} \stackrel{2x-1}{< 1} \stackrel{2x^2+x-1=(2x-1)(x+1)=0}{x_1=-1, x_2=\frac{1}{2}} \stackrel{2x-1}{< 1} \\ f'_y = \frac{y}{x^2} \stackrel{2x-1}{< 1} \stackrel{2x^2+x-1=(2x-1)(x+1)=0}{x_1=-1, x_2=\frac{1}{2}} \stackrel{2x-1}{< 1} \\ f''_{xx} = \frac{-2x^2 - 2x + 3 + y^2}{x^4} \end{cases}$$

$$f''_{yy} = \frac{1}{x^2} \quad f''_{xy} = \frac{-2y}{x^3}$$

$$(-1, 0) \quad A=3 \quad B=0 \quad C=1 \quad \Delta > 0 \quad A > 0 \quad \text{极小值} \quad f(-1, 0) = 2$$

$$(\frac{1}{2}, 0) \quad A=\frac{1}{2} \quad B=0 \quad C=4 \quad \Delta > 0 \quad A > 0 \quad \text{极小值} \quad f(\frac{1}{2}, 0) = \frac{1}{2} - 4\ln 2$$

例 13.22

例 13.22 已知函数 $z = z(x, y)$ 由方程 $(x^2 + y^2)z + \ln z + 2(x + y + 1) = 0$ 确定, 求 $z = z(x, y)$ 的极值.

$$f(x, y, z) = (x^2 + y^2)z + \ln z + 2(x + y + 1) = 0$$

$$f'_x = 2xz + 2$$

$$f'_y = 2yz + 2$$

$$f'_z = x^2 + y^2 + \frac{1}{z}$$

$$\begin{cases} \frac{\partial z}{\partial x} = -\frac{f'_x}{f'_z} = -\frac{2xz+2}{x^2+y^2+\frac{1}{z}} \stackrel{2x-1}{< 1} \stackrel{2x^2+x-1=(2x-1)(x+1)=0}{x_1=-1, x_2=\frac{1}{2}} \stackrel{2x-1}{< 1} \\ \frac{\partial z}{\partial y} = -\frac{f'_y}{f'_z} = -\frac{2yz+2}{x^2+y^2+\frac{1}{z}} \stackrel{2x-1}{< 1} \stackrel{2x^2+x-1=(2x-1)(x+1)=0}{x_1=-1, x_2=\frac{1}{2}} \stackrel{2x-1}{< 1} \end{cases} \Rightarrow \begin{cases} xz = -1 \\ yz = -1 \end{cases} \Rightarrow \begin{cases} x = y = -\frac{1}{z} \\ -\frac{4}{z^2} - 2 \end{cases}$$

$$\begin{aligned} \frac{2}{z} + \ln z + 2(\frac{2}{z} + 1) &= \ln z - \frac{2}{z} + 2 = 0 \\ \Rightarrow z &= 1 \\ x &= y = -1 \end{aligned}$$

$$G(x, y, z) = \frac{\partial z}{\partial x} + \frac{2xz+2}{x^2+y^2+\frac{1}{z}} = 0$$

$$G'_x = \frac{\partial^2 z}{\partial x^2} + \frac{2z(x^2+y^2+\frac{1}{z}) - (2xz+2) \cdot 2x}{(x^2+y^2+\frac{1}{z})^2} = 0 \Rightarrow A = \frac{\partial^2 z}{\partial x^2} = \frac{2zx^2 - 2zy^2 - 2 + 4x}{(x^2+y^2+\frac{1}{z})^2} = \frac{-(\frac{4}{z}+2)}{(x^2+y^2+\frac{1}{z})^2} = \frac{-6}{9} = -\frac{2}{3}$$

$$G'_y = \frac{\partial^2 z}{\partial x \partial y} + \frac{-(2xz+2) \cdot 2y}{(x^2+y^2+\frac{1}{z})^2} = 0 \Rightarrow B = \frac{\partial^2 z}{\partial x \partial y} = \frac{(2xz+2) \cdot 2y}{(x^2+y^2+\frac{1}{z})^2} \Big|_{xz=-1} = 0$$

$$F(x, y, z) = \frac{\partial z}{\partial y} + \frac{2yz+2}{x^2+y^2+\frac{1}{z}} = 0$$

$$F'_y = \frac{\partial^2 z}{\partial y^2} + \frac{(2yz+2)(x^2+y^2+\frac{1}{z}) - (2yz+2) \cdot 2y}{(x^2+y^2+\frac{1}{z})^2} = 0 \Rightarrow C = \frac{\partial^2 z}{\partial y^2} = \frac{2zy^2 - 2zx^2 - 2 + 4y}{(x^2+y^2+\frac{1}{z})^2} = -\frac{2}{3}$$

$$\Delta = AC - B^2 > 0 \quad A < 0 \quad \text{极大值} \quad z(-1, -1) = 1$$

例 13.23

例 13.23 求函数 $u = x^2 + y^2 + z^2$ 在约束条件 $z = x^2 + y^2$ 和 $x + y + z = 4$ 下的最大值与最小值.

$$F(x, y, z, \mu, \lambda) = x^2 + y^2 + z^2 + \mu(x^2 + y^2 - z) + \lambda(x + y + z - 4) = 0$$

$$F'_x = 2x + 2\mu x + \lambda = 0$$

$$F'_y = 2y + 2\mu y + \lambda = 0$$

$$F'_z = 2z - \mu + \lambda = 0$$

$$F'_\mu = x^2 + y^2 - z = 0$$

$$F'_\lambda = x + y + z - 4 = 0$$

$$\begin{cases} z = x^2 \\ z = 2 - x \end{cases}$$

$$\begin{cases} x^2 + x - 2 = 0 \\ (x-1)(x+2) = 0 \end{cases}$$

$$\begin{cases} x_1 = y_1 = 1 \\ x_2 = y_2 = -2 \end{cases}$$

$$\begin{cases} z_1 = 2 \\ z_2 = 8 \end{cases}$$

$$f(1, 1, 2) = 6 \leftarrow \min$$

$$f(-2, -2, 8) = 72 \leftarrow \max$$

例 13.24

例 13.24 求曲线 $x^3 - xy + y^3 = 1 (x \geq 0, y \geq 0)$ 上的点到坐标原点的最长距离与最短距离.

距离.

$$u = x^2 + y^2$$

$$F(x, y, \mu) = x^2 + y^2 + \mu(x^3 - xy + y^3 - 1)$$

$$F'_x = 2x + \mu(3x^2 - y) = 0$$

$$F'_y = 2y + \mu(3y^2 - x) = 0$$

$$F'_\mu = x^3 - xy + y^3 - 1 = 0 \quad (x+y)(x^2 - xy + y^2) = 1$$

$$x=y \text{ 时 } 2x^3 - x^2 - 1 = 0 \Rightarrow x=1 \quad y=1 \quad u=2 \Rightarrow D=\sqrt{u}=2$$

$$d_{\max} = \sqrt{2}$$



$$\frac{2x}{y-3x^2} = \frac{2y}{x-3y^2}$$

$$2x^2 - 6xy^2 = 2y^2 - 6yx^2$$

$$x^2 - 3xy^2 = y^2 - 3yx^2$$

$$(x-y)(x+y) = 3yx(y-x)$$

$$(x-y)(x+y+3yx) = 0$$

当 $x+y+3yx=0$ 时 $\Rightarrow x, y \geq 0$, 舍去!

$$\begin{cases} x=0 \text{ 时 } y=1 \\ y=0 \text{ 时 } x=0 \end{cases} \quad f(0,1) = f(1,0) = 1 \quad d_{\min} = 1$$

例 13.25

例 13.25 在椭圆 $x^2 + 4y^2 = 4$ 上求一点, 使其到直线 $2x + 3y - 6 = 0$ 的距离最短.

$$\frac{x^2}{4} + y^2 = 1$$

$$d = \frac{|2x+3y-6|}{\sqrt{2^2+3^2}} = \frac{|2x+3y-6|}{\sqrt{13}}$$



$$u = (2x+3y-6)^2$$

$$F(x, y, \mu) = (2x+3y-6)^2 + \mu(x^2 + 4y^2 - 4) = 0$$

$$F'_x = 4(2x+3y-6) + 2\mu x = 0$$

$$F'_y = 6(2x+3y-6) + 8\mu y = 0$$

$$F'_\mu = x^2 + 4y^2 - 4 = 0$$

$$\text{解之得: } \begin{cases} x = \frac{8}{\sqrt{13}} \\ y = \frac{3}{\sqrt{13}} \end{cases} \quad \begin{cases} x = -\frac{8}{\sqrt{13}} \\ y = -\frac{3}{\sqrt{13}} \end{cases}$$

$$\rightarrow u_{\min} = \frac{|\frac{16}{\sqrt{13}} + \frac{9}{\sqrt{13}} - 6|}{\sqrt{13}} = \frac{1}{\sqrt{13}}$$

$$\frac{2(2x+3y-6)}{4} = \frac{2(2x+3y-6)}{4} \Rightarrow \frac{2x+3y-6}{2} = \frac{2x+3y-6}{2}$$

$$y = \frac{3}{8}x$$

$$\frac{27}{16}k^2 = x^2 + 4\frac{9}{64}x^2 = 4$$

$$x = \frac{8}{\sqrt{13}}$$

例 13.27



例 13.27 已知 $\frac{\partial z}{\partial x} = -\sin y + \frac{1}{1-xy}$, 且 $z(0, y) = 2\sin y + y^2$, 求函数 $z(x, y)$.

$$z = -(\sin y) \cdot x - \frac{\ln(1-xy)}{y} + \varphi(y)$$

$$z(0, y) = 0 - \frac{\ln 1}{y} + \varphi(y) = \varphi(y) = 2\sin y + y^2$$

$$\Rightarrow z(x, y) = -x \sin y - \frac{\ln(1-xy)}{y} + 2\sin y + y^2$$

例 13.28 ~~☆☆~~ 预测

例 13.28 已知二元函数 $z = f(x, y)$ 可微, 两个偏增量

$$\Delta_x z = (2 + 3x^2 y^2) \Delta x + 3xy^2 (\Delta x)^2 + y^2 (\Delta x)^3, \Delta_y z = 2x^3 y \Delta y + x^3 (\Delta y)^2,$$

且 $f(0, 0) = 1$, 求 $f(x, y)$.

$$\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} = f'_x(x, y)$$

$$\frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} = f'_x(x, y) + o$$

$$\Delta_x z = \frac{\partial z}{\partial x} \Delta x + o(\Delta x)$$

$$\Rightarrow \begin{cases} \frac{\partial z}{\partial x} = 2 + 3x^2 y^2 \\ \frac{\partial z}{\partial y} = 2x^3 y \end{cases} \quad \begin{matrix} z = 2x + x^3 y^2 + \varphi(y) \\ \downarrow \\ \frac{\partial z}{\partial y} = 2x^3 y + \varphi'(y) \\ \Rightarrow \varphi'(y) = 0 \quad \varphi(y) = C \end{matrix} \quad \Rightarrow z = 2x + x^3 y^2 + 1$$



例 13.29 ~~☆☆~~

例 13.29 设函数 $f(u)$ 具有二阶连续导数, $z = f(e^x \cos y)$ 满足

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (4z + e^x \cos y) e^{2x}.$$

若 $f(0) = 0, f'(0) = 0$, 求 $f(u)$ 的表达式.

$$z = f(e^x \cos y)$$

$$\begin{cases} \frac{\partial z}{\partial x} = f' \cdot e^x \cos y \\ \frac{\partial z}{\partial y} = f' \cdot (-e^x \sin y) \end{cases}$$

$$\begin{cases} \frac{\partial^2 z}{\partial x^2} = f'' (e^x \cos y)^2 + f' e^x \cos y \\ \frac{\partial^2 z}{\partial y^2} = f'' (e^x \sin y)^2 - f' e^x \cos y \end{cases} \quad \begin{matrix} f'' \cdot e^{2x} = (4z + e^x \cos y) e^{2x} \\ f'' = 4z + e^x \cos y \\ \text{设 } u = e^x \cos y \\ f''(u) = 4f(u) + u \end{matrix}$$

$$f''(u) - 4f(u) - u = 0$$

$$\rightarrow f(u) = C_1 e^{2u} + C_2 e^{-2u} - \frac{u}{4} \quad \Rightarrow f(u) = \frac{1}{16} e^{2u} - \frac{1}{16} e^{-2u} - \frac{u}{4} \quad *$$

$$y'' - 4y = x$$

$$\lambda^2 - 4 = 0 \Rightarrow \lambda = \pm 2$$

$$y = C_1 e^{2x} + C_2 e^{-2x}$$

$$y^* = (C_3 + C_4 x) \cdot e^0 \cdot x^0 = C_3 + C_4 x$$

$$0 - 4(C_3 + C_4 x) = x \Rightarrow C_3 = 0 \quad C_4 = -\frac{1}{4}$$

$$\Rightarrow y^* = -\frac{x}{4}$$

$$y = C_1 e^{2x} + C_2 e^{-2x} - \frac{x}{4}$$



例 13.30

例 13.30 设 $f(x, y)$ 是一阶偏导数连续的正值函数, 满足 $f'_x(x, y) + f(x, y) = 0$,
又 $f'_y(0, y) = \tan y, f(0, 0) = 1$, 求 $f(x, y)$.



$$\begin{aligned} f'_x = -f &\Rightarrow \frac{f'_x}{f} = -1 \xrightarrow{\text{积分}} \ln f = -x + \varphi(y) \\ f(x, y) &= e^{-x} \cdot e^{\varphi(y)} \\ f'_y(0, y) &= (e^{\varphi(y)})'_y = \tan y \\ e^{\varphi(y)} &= \int \tan y \, dy = -\ln|\cos y| + C \\ \text{令 } y=0 \quad f(0, 0) = 1 = e^{\varphi(0)} &\Rightarrow \ln|\cos 0| + C = 1 \quad \underline{C=1} \\ \varphi(y) &= \ln(1 + \ln|\cos y|) \\ f(x, y) &= e^{-x} \cdot (1 - \ln|\cos y|) \quad \star \end{aligned}$$

例 13.31

例 13.31 设函数 $u = f(x, y)$ 具有二阶连续偏导数, 且满足等式

$$4 \frac{\partial^2 u}{\partial x^2} + 12 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial^2 u}{\partial y^2} = 0.$$

确定 a, b 的值, 使等式在变换 $\xi = x + ay, \eta = x + by$ 下简化为 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$.



$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \\ \checkmark \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial y} = a \frac{\partial u}{\partial \xi} + b \frac{\partial u}{\partial \eta} \\ \checkmark \frac{\partial^2 u}{\partial y^2} &= a^2 \frac{\partial^2 u}{\partial \xi^2} + b^2 \frac{\partial^2 u}{\partial \eta^2} + 2ab \frac{\partial^2 u}{\partial \xi \partial \eta} \\ \checkmark \frac{\partial^2 u}{\partial x \partial y} &= a \frac{\partial^2 u}{\partial \xi^2} + b \frac{\partial^2 u}{\partial \eta^2} + (a+b) \frac{\partial^2 u}{\partial \xi \partial \eta} \end{aligned}$$

$$\begin{aligned} 4 \frac{\partial^2 u}{\partial \xi^2} + 12 \frac{\partial^2 u}{\partial \xi \partial \eta} + 5 \frac{\partial^2 u}{\partial \eta^2} &= 0 \\ \Rightarrow \begin{cases} 4 + 5a^2 + 12a = 0 \\ 4 + 5b^2 + 12b \neq 0 \\ 8 + 24ab + 5(a+b) = 0 \end{cases} \\ \begin{matrix} 5a & 2 \\ a & 2 \end{matrix} \begin{cases} a = -\frac{2}{5} & b = -2 \\ a = -2 & b = -\frac{2}{5} \end{cases} \end{aligned}$$

习 13.1

13.1 如果函数 $f(x, y)$ 在点 $(0, 0)$ 处连续, 那么下列命题正确的是 ().

- (A) 若极限 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y)}{|x| + |y|}$ 存在, 则 $f(x, y)$ 在点 $(0, 0)$ 处可微
- (B) 若极限 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y)}{x^2 + y^2}$ 存在, 则 $f(x, y)$ 在点 $(0, 0)$ 处可微
- (C) 若 $f(x, y)$ 在点 $(0, 0)$ 处可微, 则极限 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y)}{|x| + |y|}$ 存在
- (D) 若 $f(x, y)$ 在点 $(0, 0)$ 处可微, 则极限 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y)}{x^2 + y^2}$ 存在

$$\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y)}{x^2 + y^2} = \lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y)}{\underbrace{x^2 + y^2}_0} \cdot \underbrace{\frac{1}{x^2 + y^2}}_{\infty}$$

↓
可微

习13.2

13.2 设 $F(x, y)$ 在点 (x_0, y_0) 的某邻域内有二阶连续偏导数, 且 $A < 0$

$$F(x_0, y_0) = 0, F'_x(x_0, y_0) = 0, F'_y(x_0, y_0) > 0, F''_{xx}(x_0, y_0) < 0,$$

则由方程 $F(x, y) = 0$ 确定的隐函数 $y = y(x)$ 在点 $x = x_0$ 处 (C).

(A) 取得极小值

(B) 取得极大值

(X) 不取得极值

(D) 不能确定是否取得极值

$$y''_{xx} = \frac{F''_{xx} F'_y - F'_x F''_{xy}}{(F'_y)^2} > 0$$

$$y'_x = -\frac{F'_x}{F'_y} = 0$$

习13.3

13.3 设 $u(x, y)$ 在平面有界闭区域 D 上具有二阶连续偏导数, 且

$$\frac{\partial^2 u}{\partial x \partial y} \neq 0, \frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} = 0,$$

则 $u(x, y)$ 的 (B).

(A) 最大值点和最小值点必定都在 D 的内部

(B) 最大值点和最小值点必定都在 D 的边界上

(C) 最大值点在 D 的内部, 最小值点在 D 的边界上

(D) 最小值点在 D 的内部, 最大值点在 D 的边界上

$$AC - B^2 < 0$$

习13.4

13.4 当 $x \geq 0, y \geq 0$ 时, $x^2 + y^2 \leq ke^{x+y}$ 恒成立, 则 k 的取值范围是 _____.

$$F(x, y) = x^2 + y^2 - ke^{x+y} \leq 0 \quad A$$

$$F'_x = 2x - ke^{x+y} \quad F''_{xx} = 2 - ke^{x+y} \quad F''_{xy} = -ke^{x+y} \quad B$$

$$F'_y = 2y - ke^{x+y} \quad F''_{yy} = 2 - ke^{x+y}$$

$$AC - B^2 = (2 - ke^{x+y})^2 - (ke^{x+y})^2 = 4 - 2ke^{x+y} = 4 - 2x = 4 - 4y$$

$$\text{极值点} \quad 2x = 2y = ke^{x+y}$$

$$k = \frac{2x}{e^{2x}} = \frac{2y}{e^{2y}}$$

$$\Rightarrow F(x, y) = x^2 - 2x + y^2$$

$$F(x, y) = \frac{x^2 + y^2}{e^{x+y}} \leq k$$

$$\text{内部} \quad F'_x = \frac{2x - (x^2 + y^2)}{e^{(x+y)}} = 0$$

$$F''_{xx} = \frac{(2-2x) - (2x-x^2-y^2)}{e^{x+y}}$$

$$F''_{xy} = \frac{-2y - (2x-x^2-y^2)}{e^{x+y}}$$

$$B(1,1) = -2$$

$$B(0,0) = 0$$

$$F''_{yy} = \frac{(2-2y) - (2y-x^2-y^2)}{e^{x+y}}$$

$$C(1,1) = 0$$

$$C(0,0) = 2$$

$$\Rightarrow x = y = \frac{x^2 + y^2}{2} \Rightarrow \underbrace{(1,1)}_x \quad \underbrace{(0,0)}_{\text{边界}}$$

$$x=0 \text{ 时} \Rightarrow (0,0) \quad (0,2)$$

$$y=0 \text{ 时} \Rightarrow (0,0) \quad (2,0)$$

$$\Rightarrow \max = 4e^{-2}$$

习13.5

13.5 设函数 $f(x, y)$ 的一阶偏导数连续, 在点 $(1, 0)$ 的某邻域内有

$$f(x, y) = 1 - x - 2y + o(\sqrt{(x-1)^2 + y^2})$$

成立. 记 $z(x, y) = f(e^x, x+y)$, 则 $d[z(x, y)]|_{(0,0)} = -2dx - 3dy$

$$z(x, y) = 1 - e^y - 2(x+y) + o(\sqrt{(e^y-1)^2 + (x+y)^2})$$

$$(-2) \quad e^y - 2 \Big|_0 = (-3)$$

习13.6

13.6 求极限 $I = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \left(1 + \frac{1}{xy}\right)^{\frac{x^2}{x^2+y}}$, 其中 $a \neq 0$.

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \left(1 + \frac{1}{xy}\right)^{\frac{x^2}{x^2+y}} = \lim_{x \rightarrow 0} \left(1 + \frac{x}{a}\right)^{\frac{1}{x+x^2 y}} = \lim_{x \rightarrow 0} e^{\frac{1}{x+x^2 y} \cdot \frac{x}{a}} = \lim_{x \rightarrow 0} e^{\frac{1}{a} \cdot \frac{1}{1+yx}} = e^{\frac{1}{a}}$$

习 13.7

13.7 求极限 $I = \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{x^3 + e^y}{x^2 + y^2}$.

$$I = \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{x^3 + e^y}{x^2 + y^2} = f(1, 0) = 2$$

习 13.8

13.8 (1) 设 $y = \frac{1}{x(1-x)}$, 求 $\frac{d^n y}{dx^n}$;

(2) 设 $z = \frac{y^2}{x(1-x)}$, 求 $\frac{\partial^n z}{\partial x^n}$.

$$(1) \quad y = \frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x} = x^{-1} - (x-1)^{-1}$$

$$\frac{d^n y}{dx^n} = (-1)^n n! \left[x^{-(n+1)} - (x-1)^{-(n+1)} \right]$$

$$(2) \quad z = y^2 (x^{-1} - (x-1)^{-1})$$

$$\frac{\partial^n z}{\partial x^n} = (-1)^n n! \left[x^{-(n+1)} - (x-1)^{-(n+1)} \right] \cdot y^2$$

习 13.9

13.9 已知函数 $f(u, v)$ 具有二阶连续偏导数, $f(1, 1) = 2$ 是 $f(u, v)$ 的极值, $z =$

$f[x+y, f(x, y)]$, 求 $\frac{\partial^2 z}{\partial x \partial y} \Big|_{(1,1)}$.

$$\frac{\partial z}{\partial x} = f'_1(x+y, f(x, y)) + f'_2(x+y, f(x, y)) \cdot f'_x(x, y)$$

$$\frac{\partial^2 z}{\partial x \partial y} = f''_{11}(x+y, f(x, y)) + f''_{12}(x+y, f(x, y)) \cdot \underbrace{f'_y(x, y)}_0 + f''_{21}(x, y) f'_2(x+y, f(x, y)) + \underbrace{f'_x(x, y)}_0 \left[f''_{21}(x+y, f(x, y)) + f''_{22}(x+y, f(x, y)) \cdot \underbrace{f'_y(x, y)}_0 \right]$$

$$\frac{\partial^2 z}{\partial x \partial y} \Big|_{(1,1)} = f''_{11}(2, 2) + f''_{21}(1, 1) \cdot f'_2(2, 2)$$

习 13.10

13.10 设 $z = z(x, y)$ 由方程 $ax + by + cz = \varphi(x^2 + y^2 + z^2)$ 所确定, 其中 φ 可微, 且

$c - 2z\varphi' \neq 0$, 证明

$$(cy - bz) \frac{\partial z}{\partial x} + (az - cx) \frac{\partial z}{\partial y} = bx - ay.$$

$$F(x, y, z) = ax + by + cz - \varphi = 0$$

$$\begin{cases} F'_x = a - \varphi' \cdot 2x \\ F'_y = b - \varphi' \cdot 2y \\ F'_z = c - \varphi' \cdot 2z \end{cases}$$

$$\frac{\partial z}{\partial x} = - \frac{F'_x}{F'_z} = - \frac{a - 2x\varphi'}{c - 2z\varphi'}$$

$$\frac{\partial z}{\partial y} = - \frac{F'_y}{F'_z} = - \frac{b - 2y\varphi'}{c - 2z\varphi'}$$

$$(cy - bz) \frac{\partial z}{\partial x} + (az - cx) \frac{\partial z}{\partial y} = \frac{(cy - bz)(a - 2x\varphi') + (az - cx)(b - 2y\varphi')}{-(c - 2z\varphi')} = bx - ay$$

习 13.11

13.11 设 $u = f(x, y, z), \varphi(x^2, e^y, z) = 0, y = \sin x$, 其中 f, φ 具有一阶连续的偏导数,

且 $\frac{\partial \varphi}{\partial z} \neq 0$, 求 $\frac{du}{dx}$.

$$\begin{aligned}\frac{du}{dx} &= f'_1 + f'_2 \cdot \cos x + f'_3 \cdot \frac{\partial z}{\partial x} \\ &= f'_1 + f'_2 \cos x - f'_3 \frac{\varphi'_x}{\varphi'_z} \\ &= f'_1 + f'_2 \cos x - f'_3 \frac{\varphi'_1 \cdot 2x + \varphi'_2 \cdot e^y \cos x}{\varphi'_3}\end{aligned}$$

习 13.12

13.12 求函数 $z = 3axy - x^3 - y^3$ 的极值, 其中 $a \neq 0$.

$$\begin{aligned}\frac{\partial z}{\partial x} &= 3ay - 3x^2 = 0 & \frac{\partial^2 z}{\partial x^2} &= -6x = A & AC-B^2 &= 3bxy - 9a^2 \\ \frac{\partial^2 z}{\partial x \partial y} &= 3a = B & \frac{\partial^2 z}{\partial x \partial y} &= 3a = B & AC-B^2 &= -9a^2 < 0 & \text{不是极值} \\ \frac{\partial z}{\partial y} &= 3ax - 3y^2 = 0 & \frac{\partial^2 z}{\partial y^2} &= -6y = C & AC-B^2 &= 27a^2 > 0 \\ ay &= x^2 & ax &= y^2 & A(a, a) &= -6a \\ y &= \frac{x^2}{a} & ax &= \frac{x^4}{a^2} \Rightarrow a^2 = x^2 & a > 0 \text{ 时 } (a, a) & \text{ 极小 } z(a, a) = a^3 \\ & & \begin{cases} x=0, \\ y=0 \end{cases} & \begin{cases} x=a \\ y=a \end{cases} & a < 0 \text{ 时 } (a, a) & \text{ 极大} \\ & & P_1 & P_2 & & \end{aligned}$$

习 13.13

13.13 求函数 $z = x^4 + y^4 - x^2 - 2xy - y^2$ 的极值.

$$\begin{aligned}\frac{\partial z}{\partial x} &= 4x^3 - 2x - 2y \stackrel{?}{=} 0 & \frac{\partial^2 z}{\partial x^2} &= 12x^2 - 2 = A \\ \frac{\partial z}{\partial y} &= 4y^3 - 2y - 2x \stackrel{?}{=} 0 & \frac{\partial^2 z}{\partial x \partial y} &= -2 = B \\ \Rightarrow 2x^3 &= 2y^3 = x+y & \frac{\partial^2 z}{\partial y^2} &= 12y^2 - 2 = C \\ \begin{cases} x=0 \\ y=0 \end{cases} & \begin{cases} x=1 \\ y=1 \end{cases} & \begin{cases} x=-1 \\ y=-1 \end{cases} & (x, y) = (0, 0) \text{ 不是极值点} \\ A < 0 & A > 0 & A > 0 & (x, y) = (1, 1), (-1, -1) \text{ 是极小值点} \\ AC-B^2 &= -4 & 96 > 0 & 96 > 0 & z = -2\end{aligned}$$

习 13.14

13.14 求二元函数 $z = f(x, y) = x^2 y(4 - x - y)$ 在由直线 $x + y = 6$, x 轴和 y 轴所围

成的闭区域 D 上的极值、最大值与最小值.

$$\begin{aligned}\textcircled{1} \quad u &= x^2 y(4 - x - y) - \mu(x + y - 6) \\ u'_x &= 2xy(4 - x - y) - x^2 y - \mu = 0 & 8xy - 3x^2 y - 2xy^2 &= xy(8 - 3x - 2y) = \mu \\ u'_y &= x^2(4 - x - y) - x^2 y - \mu = 0 & x^2(4 - x - 2y) &= \mu \\ u'_\mu &= x + y - 6 = 0 & \Rightarrow x(4 - x - 2y) &= xy(8 - 3x - 2y) \\ x=0 \text{ 时 } & y=6 & x \neq 0 \text{ 时 } & x(4 - 6 - y) = y(8 - x - 2y) \\ & & & x(2 + y) = y(4 + x) \\ & & & x(2 + 6 - x) = (6 - x)(4 + x) \\ & & & 8x - x^2 = 24 + 2x - x^2 \quad 6x = 24 \quad \begin{cases} y=2 \\ x=4 \end{cases}\end{aligned}$$

$$f(0,6) = 0 \quad f(4,2) = 32(4-4-2) = -64$$

$$\textcircled{2} \quad z'_x = 2xy(4-x-y) - x^2y = xy(8-x-2y) \stackrel{?}{=} 0$$

$$z'_y = x^2(4-x-y) - x^2y = x^2(4-x-2y) \stackrel{?}{=} 0$$

$$A \quad z''_{xx} = y(8-3x-2y) - 2xy = y(8-6x-2y)$$

$$C \quad z''_{yy} = -2x^2$$

$$B \quad z''_{xy} = x(8-3x-4y)$$

$$(0,k) \quad A = k(8-2k) \quad AC-B^2 = 0$$

$$(4,0) \quad A = 0$$

$$(2,1) \quad A = 8-12-2 < 0 \quad AC-B^2 = (-6)(8) - 4(8-6-4)^2 > 0$$

$$\text{极大值 } f(2,1) = 4$$

$$\text{极小值 } f(4,2) = -64$$

习13.15

13.15 已知函数 $f(x,y)$ 满足

$$f''_{xy}(x,y) = 2(y+1)e^x, f'_x(x,0) = (x+1)e^x, f(0,y) = y^2 + 2y,$$

求 $f(x,y)$ 的极值.

$$f'_x(x,y) = \int 2(y+1)e^x dy = (y^2+2y)e^x + \varphi(x) \Big|_{y=0} = \varphi(x) = (x+1)e^x$$

$$\rightarrow f'_x(x,y) = (y^2+2y)e^x + (x+1)e^x$$

$$\int (y^2+2y+1)e^x + xe^x dx = (y^2+2y+1)e^x + (x-1)e^x + \psi(y) \Big|_{x=0} = \frac{y^2+2y+1-1+\psi(y)}{1} = \frac{y^2+2y}{1} \Rightarrow \psi(y) = 0$$

$$\Rightarrow f(x,y) = (y^2+2y+x)e^x$$

$$f'_x = (y^2+2y+x+1)e^x \stackrel{?}{=} 0 \Rightarrow x=0 \quad (0,-1)$$

$$f'_y = (2y+2)e^x \stackrel{?}{=} 0 \Rightarrow y=-1$$

$$f''_{xx} = (y^2+2y+x+2)e^x = A \quad A > 0$$

$$f''_{xy} = (2y+2)e^x = B \quad AC-B^2 = 1 \times 2 - 0 > 0$$

$$f''_{yy} = 2e^x = C \quad f(0,-1) \text{ 是极小值. } -1$$

$$f''_{xx} = 2e^x = C$$

习13.17

13.17 设函数 $u = f(\ln \sqrt{x^2+y^2})$ 有二阶连续偏导数, 满足 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (x^2+y^2)^{\frac{3}{2}}$, 且

极限 $\lim_{x \rightarrow 0} \frac{\int_0^1 f(xt) dt}{x} = -1$, 求函数 $f(x)$ 的表达式.

$$f(0) = 0$$

$$\lim_{x \rightarrow 0} \frac{\int_0^1 f(xt) dt}{x} = -1 \quad \begin{matrix} xt=u \\ du=xdt \end{matrix} \quad \lim_{\substack{u \rightarrow 0 \\ x \rightarrow 0}} \frac{\int_0^x f(u) \frac{du}{x}}{x-0} = \lim_{u \rightarrow 0} \frac{\int_0^x f(u) du}{x^2} = \lim_{x \rightarrow 0} \frac{f(x)}{2x} = -1 \Rightarrow f'(0) = -2$$

$$\frac{\partial u}{\partial x} = f' \cdot \frac{x}{x^2+y^2} \quad \frac{\partial^2 u}{\partial x^2} = f'' \frac{x^2}{(x^2+y^2)^2} + f' \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial u}{\partial y} = f' \cdot \frac{y}{x^2+y^2} \quad \frac{\partial^2 u}{\partial y^2} = f'' \frac{y^2}{(x^2+y^2)^2} + f' \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$f(x) = \frac{e^{x^2}}{25} - \frac{6}{5}x - \frac{1}{25}$$

$$f'(x) = \frac{1}{5}e^{x^2} - \frac{6}{5}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f'' \cdot \frac{1}{x^2+y^2} = (x^2+y^2)^{\frac{3}{2}} \Rightarrow f'' = (x^2+y^2)^{\frac{5}{2}}$$

$$f''(z) = e^{z^2} \quad f'(z) = \frac{1}{5}e^{z^2} + k_1 \quad f'(0) = \frac{1}{5}e^0 + k_1 = -2 \Rightarrow k_1 = -\frac{11}{5}$$

例 13.18

13.18 设函数 $f(x, y)$ 可微, $\frac{\partial f}{\partial x} = -f$, $f(0, \frac{\pi}{2}) = 1$, 且满足

$$\lim_{n \rightarrow \infty} \left[\frac{f(0, y + \frac{1}{n})}{f(0, y)} \right]^n = e^{\cot y}, \quad (1+x)$$

求 $f(x, y)$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left[\frac{f(0, y + \frac{1}{n}) - f(0, y)}{f(0, y)} + 1 \right]^n &= \lim_{n \rightarrow \infty} \left[\frac{\frac{f(0, y + \frac{1}{n}) - f(0, y)}{\frac{1}{n}}}{f(0, y)} + 1 \right]^n \\ &= \lim_{n \rightarrow \infty} \left[\frac{f'_y(0, y)}{n f(0, y)} + 1 \right]^n = e^{\frac{f'_y(0, y)}{f(0, y)}} = e^{\cot y} \Rightarrow \frac{f'_y(0, y)}{f(0, y)} = \cot y \end{aligned}$$

$$\begin{cases} y' = \cot x \cdot y \\ y' - \cot x y = 0 \end{cases} \quad y = e^{\int \cot x dx} \left[\int e^{-\int \cot x dx} \cdot 0 dx + C \right] = C e^{\int \frac{\cos x}{\sin x} dx} = C e^{\ln |\sin x|}$$

$$\Rightarrow f(0, y) = \varphi(x) \cdot e^{\ln |\sin y|} \Big|_{x=0} = \varphi(x) \Big|_{x=0} \cdot |\sin y|$$

$$f(x, y) = \varphi(x) |\sin y|$$

$$\frac{\partial f}{\partial x} = \varphi'(x) |\sin y| = -\varphi(x) |\sin y|$$

$$\Rightarrow \varphi(x) = k e^{-x}$$

$$\Rightarrow f(x, y) = k e^{-x} |\sin y|$$

$$f(0, \frac{\pi}{2}) = k = 1$$

$$\Rightarrow f(x, y) = e^{-x} \sin y$$

例 13.19

13.19 设 $z = z(x, y)$ 有二阶连续偏导数, 用变换 $u = x - 2y, v = x + ay$ 可把方程 $6 \frac{\partial^2 z}{\partial x^2} +$

$\frac{\partial^2 z}{\partial x \partial y} - \frac{\partial^2 z}{\partial y^2} = 0$ 简化为 $\frac{\partial^2 z}{\partial u \partial v} = 0$, 求常数 a .

$$\begin{cases} u = x - 2y \\ v = x + ay \end{cases} \quad \begin{cases} y = \frac{1}{a+2} (v-u) \\ x = \frac{1}{a+2} (au - 2v) \end{cases}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = -2 \frac{\partial z}{\partial u} + a \frac{\partial z}{\partial v}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} + 2 \frac{\partial^2 z}{\partial u \partial v}$$

$$\frac{\partial^2 z}{\partial y^2} = 4 \frac{\partial^2 z}{\partial u^2} + a \frac{\partial^2 z}{\partial v^2} - 4a \frac{\partial^2 z}{\partial u \partial v}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -2 \frac{\partial^2 z}{\partial u^2} + a \frac{\partial^2 z}{\partial v^2} + (a-2) \frac{\partial^2 z}{\partial u \partial v}$$

$$6 \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - \frac{\partial^2 z}{\partial y^2}$$

$$= (6-2-4) \frac{\partial^2 z}{\partial u^2} - (6+a-a^2) \frac{\partial^2 z}{\partial v^2} + (12+a-2+4a) \frac{\partial^2 z}{\partial u \partial v}$$

$$\Rightarrow a^2 - a - 6 = 0 \quad a^2 - a - 3 \quad a = 3 \text{ 或 } a = -2 \quad \sqrt{a+10} \neq 0 \Rightarrow a \neq -2 \Rightarrow a = 3$$

1000. 13. 1. 1

1. 设 $F(x, y) = \left(\frac{x}{y}\right)^{\frac{1}{x-y}}$, 其中 $x \neq y$, 且 $xy > 0$. 又设

$$f(x) = \begin{cases} \lim_{y \rightarrow x} F(x, y), & x \neq 0, \\ e, & x = 0, \end{cases}$$

则点 $x = 0$ 为 $f(x)$ 的 (D).

(A) 连续点

(B) 可去间断点

(C) 跳跃间断点

(D) 无穷间断点

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left(\frac{x}{y}\right)^{\frac{1}{x-y}} = \lim_{(x,y) \rightarrow (0,0)} e^{\frac{1}{x-y}(\frac{x}{y}-1)} = \lim_{(x,y) \rightarrow (0,0)} e^{\frac{1}{y}} = \infty$$

1000. 13. 1. 2

2. 设 $f(x, y) = \frac{x}{y^2} e^{-(\frac{x}{y})^2}$, $y \neq 0$, 记 $I_1 = \lim_{y \rightarrow 0} \left[\int_0^1 f(x, y) dx \right]$, $I_2 = \int_0^1 \left[\lim_{y \rightarrow 0} f(x, y) \right] dx$, 则 (A).

(A) $I_1 > I_2$

(B) $I_1 < I_2$

(C) $I_1 = I_2$

(D) I_1 与 I_2 的大小关系不确定

$$I_1 = \lim_{y \rightarrow 0} \left[\int_0^1 f(x, y) dx \right]$$

$$= \lim_{y \rightarrow 0} \frac{1}{2} \int_0^1 e^{-(\frac{x}{y})^2} d(\frac{x}{y})^2$$

$$= \lim_{y \rightarrow 0} \frac{1}{2} e^{-(\frac{x}{y})^2} \Big|_0^1 = \frac{1}{2}$$

$$I_2 = \int_0^1 \lim_{y \rightarrow 0} \frac{x e^{-(\frac{x}{y})^2}}{y^2} dx$$

$$\frac{1}{y} = z \quad \int_0^1 \lim_{z \rightarrow \infty} \frac{x \cdot z^2}{e^{x^2 z^2}} dx$$

$$= 0$$

$$x e^{-(\frac{x}{y})^2} \cdot 2x \cdot y^{-3} = \frac{x^2}{y^3} e^{-\frac{x^2}{y^2}} = -x y^{-2}$$

1000. 13. 1. 3

3. 设 $f(x, y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x}, & xy \neq 0, \\ 0, & xy = 0, \end{cases}$ 记 $I_1 = \lim_{x \rightarrow 0} [\lim_{y \rightarrow 0} f(x, y)]$, $I_2 = \lim_{y \rightarrow 0} f(x, y)$, 则 (B).

(A) I_1 存在, I_2 不存在

(B) I_1 不存在, I_2 存在

(C) I_1 存在, I_2 存在

(D) I_1 不存在, I_2 不存在

1000. 13. 1. 4

4. 下列二元函数 $f(x, y)$ 在点 $(0, 0)$ 处可微的是 (C).

$$\times (A) f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$\times (B) f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$(C) f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{\sqrt{x^2 + y^2}}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$(D) f(x, y) = \begin{cases} \sqrt{x^2 + y^2} \sin \frac{1}{x^2 + y^2}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2) \sin \frac{1}{\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}}$$

1000.13.1.5

5. 设 $f(x, y) = e^{x+y} [x^{\frac{1}{3}}(y-1)^{\frac{1}{3}} + y^{\frac{1}{3}}(x-1)^{\frac{1}{3}}]$, 则在点 $(0, 1)$ 处的两个偏导数 $f'_x(0, 1)$ 和 $f'_y(0, 1)$ 的情况为 (C).

- (A) 两个偏导数均不存在 (B) $f'_x(0, 1)$ 不存在, $f'_y(0, 1) = \frac{4}{3}e$
 (C) $f'_x(0, 1) = \frac{e}{3}, f'_y(0, 1) = \frac{4}{3}e$ (D) $f'_x(0, 1) = \frac{e}{3}, f'_y(0, 1)$ 不存在

$$\begin{aligned} f(x, y) &= e^{x+y} \left[(x-1)^{\frac{1}{3}} + \frac{1}{3}(x-1)^{-\frac{2}{3}} \right] \\ f'_x &= e^{x+y} \left[(x-1)^{\frac{1}{3}} + \frac{1}{3}(x-1)^{-\frac{2}{3}} \right] \Big|_{x=0} = e \left[1 - \frac{2}{3} \right] = \frac{1}{3}e \\ f(0, y) &= e^y \left[y^{\frac{1}{3}} \right] \\ f'_y &= e^y \left[y^{\frac{1}{3}} + \frac{1}{3}y^{-\frac{2}{3}} \right] \Big|_{y=1} = e \left[1 + \frac{1}{3} \right] = \frac{4}{3}e \end{aligned}$$

1000.13.1.6

6. 设函数 $f(x, y)$ 连续, $f(0, 0) = 0$, 又设 $F(x, y) = |x - y|f(x, y)$, 则 $F(x, y)$ 在点 $(0, 0)$ 处 (D).

- (A) 连续, 但不可微 (B) 连续, 但偏导数不存在
 (C) 偏导数存在, 但不可微 (D) 可微

$$\textcircled{1} F(x, y) = |x - y|f(x, y)$$

$$F(0, 0) = 0 \cdot 0 = 0 \quad \text{连续} \quad |x - y|$$

$$\textcircled{2} F(x, y) - F(0, 0) = |x - y|f(x, y)$$

$$F(x, 0) = |x|f(x, 0)$$

$$\frac{\partial F}{\partial x} \Big|_{y=0} = \lim_{x \rightarrow 0} \frac{|x|f(x, 0) - 0}{x - 0} = f(0, 0) = 0$$

$$\text{同理} \quad F'_y(0, 0) = 0$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{|x - y|f(x, y) - (0 \cdot \frac{\partial F}{\partial x} + 0 \cdot \frac{\partial F}{\partial y})}{\sqrt{x^2 + y^2}} = 0$$

1000.13.1.7

7. 设函数 $f(x, y)$ 在点 $(0, 0)$ 处的某一邻域内有定义, 且 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y) - f(0, 0)}{\cos(x^2 + y^2) - 1} = 1$, 则下列结论

不正确的是 (B).

- (A) $f(x, y)$ 在点 $(0, 0)$ 处连续 (B) $f'_x(0, 0)$ 与 $f'_y(0, 0)$ 都存在但不为零
 (C) $f'_x(0, 0) = f'_y(0, 0) = 0$ (D) $f(x, y)$ 在点 $(0, 0)$ 处可微

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{\quad}{x^2 + y^2}$$

1000.13.1.8

8. 设 $f(x, y) = |x - y|\varphi(x, y)$, 其中函数 $\varphi(x, y)$ 在点 $(0, 0)$ 的某一邻域内有定义, 则函数 $f(x, y)$ 在点 $(0, 0)$ 处可微的充分必要条件是 (A).

- (A) $\lim_{(x, y) \rightarrow (0, 0)} \varphi(x, y) = 0$ (B) $\varphi'_x(0, 0)$ 与 $\varphi'_y(0, 0)$ 都存在
 (C) $\varphi(x, y)$ 在点 $(0, 0)$ 处连续 (D) $\varphi(x, y)$ 在点 $(0, 0)$ 处可微

$$\begin{cases} f(x, 0) = |x| \varphi(x, 0) \\ f(0, y) = |y| \varphi(0, y) \end{cases} \quad A$$

1000.13.1.9

9. 设 $y = f(x, t)$, 而 t 是由方程 $F(x, y, t) = 0$ 所确定的 x, y 的函数, 其中 f, F 均具有一阶连续偏导数, 则 $\frac{dy}{dx} =$ (B).

(A) $\frac{f'_x F'_t + f'_t F'_x}{F'_t}$

(B) $\frac{f'_x F'_t - f'_t F'_x}{F'_t}$

(C) $\frac{f'_x F'_t + f'_t F'_x}{f'_x F'_t + f'_t F'_x}$

(D) $\frac{f'_x F'_t - f'_t F'_x}{f'_x F'_t + f'_t F'_x}$

$$\frac{dy}{dx} = f'_x + f'_t \cdot t'_x \quad \frac{F'_t f'_x - f'_t F'_x}{F'_t}$$

$$t'_x = -\frac{F'_x}{F'_t} = -\frac{F'_x}{F'_t}$$

$$\begin{cases} F'_x dx + F'_y dy + F'_t dt = 0 & dt = \frac{F'_x dx + F'_y dy}{-F'_t} \\ dy = f'_x dx + f'_t dt & dy = f'_x dx - \frac{f'_t (F'_x dx + F'_y dy)}{F'_t} \end{cases}$$

$$\Rightarrow dy = \frac{f'_x F'_t - f'_t F'_x}{f'_t F'_y + F'_t} dx \quad D$$

1000.13.1.10

10. 设 $y = f(x)$ 是由方程 $F\left(\ln \frac{x}{y}, \frac{x^2 - y^2}{xy}\right) = 0$ 确定的函数, 其中函数 $F(u, v)$ 具有连续偏导数, 且 $F'_u \cdot F'_v > 0$, 则 $\frac{dy}{dx} =$ (A).

(A) $\frac{y}{x}$

(B) $\frac{x}{y}$

(C) $-\frac{y}{x}$

(D) $-\frac{x}{y}$

$$\left[F'_u \cdot \frac{1}{x} + F'_v \left(\frac{1}{y} + \frac{y}{x^2} \right) \right] dx + \left[F'_u \cdot \left(-\frac{1}{y} \right) + F'_v \left(-\frac{x}{y^2} - \frac{1}{x} \right) \right] dy = 0$$

$$\frac{dy}{dx} = \frac{F'_u \frac{1}{x} + F'_v \left(\frac{1}{y} + \frac{y}{x^2} \right)}{F'_u \left(-\frac{1}{y} \right) + F'_v \left(-\frac{x}{y^2} - \frac{1}{x} \right)} = \frac{F'_u y + F'_v (x + \frac{y^2}{x})}{F'_u x + F'_v (y + \frac{x^2}{y})} = \frac{F'_u + F'_v (\frac{x}{y} + \frac{y}{x})}{F'_u + F'_v (\frac{x}{y} + \frac{y}{x})} \cdot \frac{y}{x} = \frac{y}{x}$$

1000.13.1.11

11. 设 $u = xe^{-y}z^2$, 若函数 $z = z(x, y)$ 由方程 $e^{x+y-z} + xyz = 1$ 确定, 记 $a = \frac{\partial u}{\partial x} \Big|_{(1,0,1)}$; 若该方程也可确定函数 $y = y(x, z)$, 记 $b = \frac{\partial u}{\partial x} \Big|_{(1,0,1)}$, 则 ().

(A) $a = 3, b = 3$

(B) $a = 3, b = \frac{3}{2}$

(C) $a = \frac{3}{2}, b = 3$

(D) $a = \frac{3}{2}, b = \frac{3}{2}$

① $u = xe^{-y}z^2$

$$F(x, y, z) = e^{x+y-z} + xyz - 1 = 0$$

$$\frac{dz}{dx} = -\frac{F'_x}{F'_z} = \frac{e^{x+y-z} + yz}{e^{x+y-z} - xy}$$

$$\frac{\partial u}{\partial x} = e^{-y}z^2 + 2xe^{-y}z \cdot \frac{e^{x+y-z} + yz}{e^{x+y-z} - xy} \Big|_{(1,0,1)} = 1 + 2 \cdot \frac{1}{1} = 3 = a$$

② $u = xe^{-y}z^2$

$$du = e^{-y}z^2 dx - xe^{-y}z^2 dy + 2xe^{-y}z dz$$

$$\frac{du}{dx} = e^{-y}z^2 - xe^{-y}z^2 \frac{dy}{dx} + 2xe^{-y}z \frac{dz}{dx} \Big|_{(1,0,1)}$$

$$= 1 + \frac{1}{2} + 2 = \frac{5}{2}$$

$$G(x, y, z) = e^{x+y-z} + xyz - 1$$

$$\frac{dy}{dx} = -\frac{G'_x}{G'_y} = -\frac{e^{x+y-z} + yz}{e^{x+y-z} + xz} \Big|_{(1,0,1)} = -\frac{1}{2}$$

1000.13.1.12

12. 若函数 $u = xyf\left(\frac{x+y}{xy}\right)$, 其中 f 是可微函数, $f \neq 0$, 且 $x^2 \frac{\partial u}{\partial x} - y^2 \frac{\partial u}{\partial y} = G(x, y)u$, 则函数 $G(x, y) =$ (B).
- (A) $x + y$ (B) $x - y$ (C) $x^2 - y^2$ (D) $(x + y)^2$

$$u = xyf\left(\frac{x+y}{xy}\right)$$

$$\frac{\partial u}{\partial x} = yf + xyf' \frac{xy - (x+y)y}{x^2y^2} = yf - f' \frac{y}{x}$$

$$x^2 \frac{\partial u}{\partial x} = x^2 y f - xy f'$$

$$\frac{\partial u}{\partial y} = xf - f' \frac{x}{y}$$

$$y^2 \frac{\partial u}{\partial y} = y^2 x f - f' xy$$

$$\underline{f \cdot xy(x-y)} = \underline{G(x,y) \cdot xy \cdot f}$$

1000.13.1.13

13. 函数 $z = x^3 + y^3 - 3x^2 - 3y^2$ 的极小值点是 (B).

(A) (0,0) (B) (2,2) (C) (0,2) (D) (2,0)

$$\begin{aligned} \frac{\partial z}{\partial x} &= 3x^2 - 6x \stackrel{!}{=} 0 \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases} \quad \begin{cases} x=2 \\ y=0 \end{cases} \quad \begin{cases} x=0 \\ y=2 \end{cases} \quad \boxed{\begin{cases} x=2 \\ y=2 \end{cases}} \\ \frac{\partial z}{\partial y} &= 3y^2 - 6y \stackrel{!}{=} 0 \quad \begin{matrix} AC-B^2 >0 <0 <0 >0 \\ A <0 >0 <0 >0 \end{matrix} \\ \frac{\partial^2 z}{\partial x^2} &= 6x - 6 = A \quad \begin{matrix} \text{极大} \end{matrix} \quad \begin{matrix} \text{极小} \end{matrix} \\ \frac{\partial^2 z}{\partial x \partial y} &= 0 = B \\ \frac{\partial^2 z}{\partial y^2} &= 6y - 6 = C \end{aligned}$$

1000.13.1.14

14. 函数 $z = \frac{x^2}{2} + xy + \frac{y^2}{2} - 2x - 2y + 5$ (D).

- (A) 有无穷多个极小值点, 没有极大值点
(B) 有无穷多个极大值点, 没有极小值点
(C) 有无穷多个极大值点, 也有无穷多个极小值点
(D) 既没有极大值点, 也没有极小值点

$$\begin{aligned} \frac{\partial z}{\partial x} &= x+y-2 \stackrel{!}{=} 0 \Rightarrow x+y=2 \quad \frac{4+x-4x}{2} \\ \frac{\partial z}{\partial y} &= x+y-2 \stackrel{!}{=} 0 \\ A = \frac{\partial^2 z}{\partial x^2} &= 1 > 0 \quad z = \frac{x^2}{2} + x(2-x) + \frac{(2-x)^2}{2} - 4 \\ B = 1 \quad AC-B^2 &= 0 \quad -\frac{x^2}{2} + 2x - x^2 + 2 + \frac{x^2}{2} - 2x - 4 \\ C = 1 \quad &= -2 \\ z &= \frac{1}{2}(x+y)^2 - 2(x+y) + 5 = \frac{1}{2}[(x+y)-2]^2 + 3 \geq 3 \end{aligned}$$

1000.13.1.15

15. 设 $f(x, y)$ 是连续函数, 且 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{f(x, y) - f(0, 0)}{x^3 + y^3 - 3x^2 - 3y^2} = 1$, 则 (A).

- (A) $f(0, 0)$ 是 $f(x, y)$ 的极大值 (B) $f(0, 0)$ 是 $f(x, y)$ 的极小值
(C) $f(0, 0)$ 不是 $f(x, y)$ 的极值 (D) 不能确定

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{\int x^2 + y^2} \cdot \frac{\sqrt{x^2 + y^2}}{x^2(x-3) + y^2(y-3)} = 1$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{x^2 + y^2} = -3$$

15. (A) 【解析】记

$$F(x, y) = x^3 + y^3 - 3x^2 - 3y^2,$$

则有 $F'_x = 3x^2 - 6x \stackrel{\text{令}}{=} 0, F'_y = 3y^2 - 6y \stackrel{\text{令}}{=} 0$, 解得 $\begin{cases} x = 0 \\ y = 0 \end{cases}$ 是其中一个驻点 (其他驻点的讨论见强化训练第 13 题).

又由

$$A = F''_{xx} \Big|_{(0,0)} = (6x - 6) \Big|_{(0,0)} = -6,$$

$$B = F''_{xy} \Big|_{(0,0)} = 0, C = F''_{yy} \Big|_{(0,0)} = (6y - 6) \Big|_{(0,0)} = -6,$$

知 $\Delta = AC - B^2 = 36 > 0$, 又 $A < 0$, 故点 $(0, 0)$ 为 $F(x, y)$ 的极大值点, 且 $F(0, 0) = 0$, 故当 $x \rightarrow 0, y \rightarrow 0$ 时, 分母为负.

由极限的保号性知, 存在 $\delta > 0$, 当点 (x, y) 满足 $0 < \sqrt{x^2 + y^2} < \delta$ 时,

$$\frac{f(x, y) - f(0, 0)}{x^3 + y^3 - 3x^2 - 3y^2} > 0, f(x, y) < f(0, 0),$$

故 $f(0, 0)$ 为极大值, 选 (A).

1000.13.1.16

16. 已知函数 $f(x, y)$ 在点 $(0, 0)$ 处的某个邻域内连续, 且 $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y)}{(x^2 + y^2)^2} = -2$, 则 (B).

(A) 点 $(0, 0)$ 不是 $f(x, y)$ 的极值点

(B) 点 $(0, 0)$ 是 $f(x, y)$ 的极大值点

(C) 点 $(0, 0)$ 是 $f(x, y)$ 的极小值点

(D) 根据所给条件无法判定点 $(0, 0)$ 是否为 $f(x, y)$ 的极值点



1000.13.1.17

17. 设函数 $u = u(x, y)$ 在有界闭区域 D 上连续, 在 D 的内部具有连续偏导数, 且满足 $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 1 + u^2$, 则 ().

(A) $u(x, y)$ 的最大值和最小值都在 D 的边界上取得

(B) $u(x, y)$ 的最大值和最小值都在 D 的内部取得

(C) $u(x, y)$ 的最大值在 D 的内部取得, 最小值在 D 的边界上取得

(D) $u(x, y)$ 的最小值在 D 的内部取得, 最大值在 D 的边界上取得

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 1 + u^2$$

1000.13.1.18

18. 设 $f(x, y)$ 与 $G(x, y)$ 均为可微函数, 且 $G'_x(x, y) \neq 0$. 已知点 (x_0, y_0) 是 $f(x, y)$ 在约束条件 $G(x, y) = 0$ 下的一个极值点, 下列选项正确的是 (D).

(A) 若 $f'_x(x_0, y_0) = 0$, 则 $f'_y(x_0, y_0) = 0$

(B) 若 $f'_x(x_0, y_0) = 0$, 则 $f'_y(x_0, y_0) \neq 0$

(C) 若 $f'_x(x_0, y_0) \neq 0$, 则 $f'_y(x_0, y_0) = 0$

(D) 若 $f'_x(x_0, y_0) \neq 0$, 则 $f'_y(x_0, y_0) \neq 0$

$$F = f(x, y) + \varphi G(x, y) = 0 \quad f'_x = \frac{f_y}{G_y} G'_x$$

$$F'_x = f'_x + \varphi G'_x = 0$$

$$F'_y = f'_y + \varphi G'_y = 0 \Rightarrow \varphi = -\frac{f'_y}{G'_y}$$

$$F'_\varphi = G = 0 \neq 0$$

1000.13.1.19

$$z''_{11} = z''_{11}$$

19. 设函数 $z = z(x, y)$ 具有二阶连续偏导数, $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$, 且满足 $z(x, 3x) = x^2$, $z'_1(x, 3x) = x^3$,则 $z''_{12}(x, 3x) =$ ().

(A) $\frac{5}{4}x^2 - \frac{1}{12}$

(B) $\frac{5}{4}x^2 + \frac{1}{12}$

(C) $\frac{4}{5}x^2 - \frac{1}{12}$

(D) $\frac{4}{5}x^2 + \frac{1}{12}$

$$\frac{\partial z(x, 3x)}{\partial x} = \frac{z'_1(x, 3x)}{x^3} + \frac{3z'_2(x, 3x)}{2x-x^3} = 2x \quad \frac{\partial z(x, y)}{\partial y} = z'_2(x, y) \quad \frac{\partial^2 z(x, y)}{\partial y^2} = z''_{22}(x, y)$$

$$\frac{\partial^2 z(x, 3x)}{\partial x \partial x} = z''_{11}(x, 3x) + 3z''_{12}(x, 3x) + 3z''_{21}(x, 3x) + 9z''_{22}(x, 3x) = z''_{11}(x, 3x) + 9z''_{22}(x, 3x) + 6z''_{12}(x, 3x) = 2$$

$$\Rightarrow 5z''_{11}(x, 3x) + 3z''_{12}(x, 3x) = 1 \Rightarrow z''_{12}(x, 3x) = \frac{1}{3}x^2 - \frac{1}{12}$$

$$z''_{11}(x, 3x) + 3z''_{22}(x, 3x) = 1$$

1000.13.1.20

$$z'_1(x, 3x) = x^3 \Rightarrow$$

20. 设 $f(x, y) = \begin{cases} \frac{\sin(x^2 y)}{2xy}, & xy \neq 0, \\ x, & xy = 0, \end{cases}$ 则 $f'_x(0, 1) =$ _____.

$$f(x) = \begin{cases} \frac{\sin x^2}{2x}, & x \neq 0 \\ x, & x = 0 \end{cases} \quad \lim_{x \rightarrow 0} \frac{\frac{\sin x^2}{2x}}{x} = \frac{1}{2}$$

1000.13.1.21

21. 设 $f(x, y, z) = \sqrt{\frac{x^4 + y^4 + z^4}{x^2 + y^2 + z^2}}$, 则 $xf'_x(x, y, z) + yf'_y(x, y, z) + zf'_z(x, y, z) =$ _____.

$$\left(\ln f = \frac{1}{2} \ln(x^4 + y^4 + z^4) - \frac{1}{2} \ln(x^2 + y^2 + z^2) \right) \quad \frac{f'}{f} = 2 \frac{x^3 dx + y^3 dy + z^3 dz}{x^4 + y^4 + z^4} - \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2}$$

$$xf'_x = \frac{x}{2} \left(\frac{x^4 + y^4 + z^4}{x^2 + y^2 + z^2} \right)^{-\frac{1}{2}} \cdot \frac{4x^3(x^2 + y^2 + z^2) - (x^4 + y^4 + z^4) \cdot 2x}{(x^2 + y^2 + z^2)^2}$$

$$f'_x = \frac{1}{2} \left[\frac{x^4 + y^4 + z^4}{x^2 + y^2 + z^2} \right]^{-\frac{1}{2}} \cdot \frac{4x^3(x^2 + y^2 + z^2) - 2(x^4 + y^4 + z^4)(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^2} = \left[\frac{x^4 + y^4 + z^4}{x^2 + y^2 + z^2} \right]^{\frac{1}{2}}$$

1000.13.1.22

22. 设 $z = z(x, y)$ 是由 $z + e^z = xy$ 确定的二元函数, 则 $\frac{\partial^2 z}{\partial x \partial y} \Big|_{z=0} =$ _____.

$$(1+e^z) \frac{\partial z}{\partial x} = y \quad \begin{cases} 1 = xy \\ 2 \frac{\partial z}{\partial x} = y \\ 2 \frac{\partial z}{\partial y} = x \end{cases} \quad (1+e^z) \frac{\partial z}{\partial y} = x$$

$$e^z \frac{\partial z}{\partial y} \frac{\partial z}{\partial x} + (1+e^z) \frac{\partial^2 z}{\partial x \partial y} = 1$$

$$\frac{\partial z}{\partial y} \frac{\partial z}{\partial x} + 2 \frac{\partial^2 z}{\partial x \partial y} = 1$$

$$\frac{1}{4} + 2 \frac{\partial^2 z}{\partial x \partial y} = 1 \Rightarrow \frac{\partial^2 z}{\partial x \partial y} = \frac{3}{8}$$

1000.13.1.23

23. 设 $z = x \ln[(1+y^2)e^{x^2 \sin y}]$, 则 $\frac{\partial^4 z}{\partial y^2 \partial x^2} =$ _____.

$$\frac{dz}{dy} = \frac{x[2y \cdot e^{x^2 \sin y} + (1+y^2) \cdot e^{x^2 \sin y} \cdot x^2 \cos y]}{(1+y^2) e^{x^2 \sin y}} = \frac{2xy + (1+y^2)x^3 \cos y}{1+y^2} = \frac{2xy}{1+y^2} + x^3 \cos y$$

$$\frac{d^2 z}{dy dx} = \frac{2y}{1+y^2} + 3x^2 \cos y \quad \frac{d^2 z}{dy dx} = 6x \cos y \quad \frac{d^4 z}{dy^2 dx} = -6x \sin y$$

1000.13.1.24

24. 设 $z = z(x, y)$ 是由方程 $x^2 y - z = \varphi(x + y + z)$ 确定的函数, 其中 φ 可导, 且 $\varphi' \neq -1$, 则 $\frac{\partial z}{\partial x} =$ _____.

$$\begin{aligned} F &= x^2 y - z - \varphi \\ F'_x &= 2xy - \varphi' \\ F'_z &= -1 - \varphi' \end{aligned} \quad \frac{\partial z}{\partial x} = \frac{2xy - \varphi'}{1 + \varphi'}$$

1000.13.1.25

25. 设 $z = x^y, x = \sin t, y = \tan t$, 则全导数 $\frac{dz}{dt} =$ _____.

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = yx^{y-1} \cdot \cos t + x^y \ln x \cdot \sec t \\ &= (\sin t)^{\tan t} = \sin^{\tan t} (\sec t \cdot \ln \sin t + 1) \\ &= e^{t \cdot \ln s} \end{aligned}$$

1000.13.1.26

26. 已知函数 $f(x, t) = \int_0^{\frac{x}{\sqrt{at}}} e^{-u^2} du, t > 0$, 若 $a \frac{\partial^2 f}{\partial x^2} + b \frac{\partial f}{\partial t} \equiv 0, a, b$ 为常数且 $a > 0$, 则 $b =$ _____.

$$f(x, t) = \int_0^{\frac{x}{\sqrt{at}}} e^{-u^2} du, t > 0$$

$$\frac{\partial f}{\partial x} = e^{-\frac{x^2}{4at}} \cdot \frac{1}{2\sqrt{at}} \quad \frac{\partial^2 f}{\partial x^2} = e^{-\frac{x^2}{4at}} \cdot \frac{1}{2\sqrt{at}} \cdot \frac{-2x}{4at}$$

$$\frac{\partial f}{\partial t} = e^{-\frac{x^2}{4at}} \cdot \frac{-x}{4\sqrt{a}} \cdot \frac{1}{t^{\frac{3}{2}}}$$

$$\begin{cases} a \frac{\partial^2 f}{\partial x^2} = e^{-\frac{x^2}{4at}} \cdot \frac{-x}{\sqrt{a} t^{\frac{3}{2}}} \\ b \frac{\partial f}{\partial t} = e^{-\frac{x^2}{4at}} \cdot \frac{-x}{4\sqrt{a} t^{\frac{3}{2}}} = \frac{1}{4} a \frac{\partial^2 f}{\partial x^2} \end{cases} \Rightarrow \begin{cases} (\frac{1}{4}a + 1) e^{-\frac{x^2}{4at}} \frac{-x}{\sqrt{a} t^{\frac{3}{2}}} = 0 \\ \Rightarrow a = -4 \\ b = -1 \end{cases}$$

1000.13.1.27

27. 已知 $f(u)$ 可导且 $f(u) \neq 0$, 则对于 $z = \frac{y}{f(x^2 - y^2)}, xy \neq 0$, $\frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} =$ _____.

$$\begin{aligned} z &= \frac{y}{f(x^2 - y^2)} \\ \frac{1}{x} z' &= \frac{1}{x} \frac{-y f' \cdot 2x}{f^2} = \frac{-2y f'}{f^2} \\ \frac{1}{y} z' &= \frac{1}{y} \frac{f + y f' 2y}{f^2} = \frac{1}{y f} + \frac{2y f'}{f^2} \end{aligned}$$

1000.13.1.28

u v

28. 设 $g(x, y) = f(2xy, x^2 - y^2)$, 其中 $f(u, v)$ 具有二阶连续偏导数, 且 $\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} = 1$, 则

$$\frac{\partial^2 g}{\partial x^2} \Big|_{\substack{x=1 \\ y=2}} + \frac{\partial^2 g}{\partial y^2} \Big|_{\substack{x=1 \\ y=2}} = 20$$

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$$\frac{\partial g}{\partial x} = \frac{\partial f}{\partial u} \cdot 2y + \frac{\partial f}{\partial v} \cdot 2x$$

$$\frac{\partial^2 g}{\partial x^2} = \frac{\partial^2 f}{\partial u^2} \cdot 4y^2 + \frac{\partial^2 f}{\partial u \partial v} \cdot 4xy + \frac{\partial^2 f}{\partial v^2} \cdot 4x^2 + \frac{\partial^2 f}{\partial u \partial v} \cdot 4xy + 2 \frac{\partial f}{\partial v}$$

$$\frac{\partial g}{\partial y} = \frac{\partial f}{\partial u} \cdot 2x + \frac{\partial f}{\partial v} \cdot (-2y)$$

$$\frac{\partial^2 g}{\partial y^2} = \frac{\partial^2 f}{\partial u^2} \cdot 4x^2 - \frac{\partial^2 f}{\partial u \partial v} \cdot 4xy + \frac{\partial^2 f}{\partial v^2} \cdot 4y^2 - \frac{\partial^2 f}{\partial u \partial v} \cdot 4xy - 2 \frac{\partial f}{\partial v}$$

$$\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} = 4x^2 + 4y^2 = 4 + 16 = 20$$

1000.13.1.29

29. 设函数 $z = xy \ln x$, 则 $d(dz) =$ _____.

$$dz = y(\ln x + 1)dx + x \ln x dy$$

$$d(dz) = \frac{y}{x} dx^2 + (\ln x + 1) dx dy + (\ln x + 1) dy dx \\ = \frac{y}{x} dx^2 + 2(\ln x + 1) dx dy$$

1000.13.1.30

30. 设 $z = z(u, v)$ 具有二阶连续偏导数, 且 $\frac{\partial^2 z}{\partial u \partial v} = 0$, 又设 $u = x^2 - y, v = f(xy)$, 其中 f 二阶可导, 满足 $f' + xyf'' = 0$, 则 $\frac{\partial^2 z}{\partial x \partial y} =$ _____.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = 2x \cdot \frac{\partial z}{\partial u} + y \cdot f' \cdot \frac{\partial z}{\partial v}$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2x \left(\frac{\partial^2 z}{\partial u^2} \cdot (-1) \right) + \left(f' + xyf'' \right) \frac{\partial z}{\partial v} + y f' \frac{\partial^2 z}{\partial v^2} \cdot x f'$$

$$= -2x \frac{\partial^2 z}{\partial u^2} + xy (f')^2 \frac{\partial^2 z}{\partial v^2}$$

1000.13.1.31

31. 设函数 $f(u)$ 具有二阶连续导数, $F(x, y) = f\left(\frac{1}{r}\right)$, $r = \sqrt{x^2 + y^2}$, 则 $\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} =$ _____.

$$\frac{\partial F}{\partial x} = f' \cdot \frac{-1}{r^2} \cdot \frac{1}{2} (x^2 + y^2)^{-\frac{1}{2}} \cdot 2x$$

$$\frac{\partial F}{\partial x} = \frac{-x f'}{r^2 \sqrt{x^2 + y^2}}$$

$$\frac{\partial F}{\partial y} = \frac{-y f'}{r^2 \sqrt{x^2 + y^2}}$$

$$\frac{\partial F}{\partial x} = \frac{-x f'}{r^3}$$

$$\frac{\partial F}{\partial y} = \frac{-y f'}{r^3}$$

$$\frac{\partial^2 F}{\partial x^2} = \frac{[-f' + x^2 f'' (x^2 + y^2)^{-\frac{1}{2}}] r^2 \sqrt{x^2 + y^2} - f' r^2 x^2 (x^2 + y^2)^{-\frac{1}{2}}}{r^4 (x^2 + y^2)}$$

$$\frac{\partial^2 F}{\partial y^2} = \frac{[-f' + y^2 f'' (x^2 + y^2)^{-\frac{1}{2}}] r^2 \sqrt{x^2 + y^2} - f' r^2 y^2 (x^2 + y^2)^{-\frac{1}{2}}}{r^4 (x^2 + y^2)}$$

$$\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} = \frac{[-2f' + f'' (x^2 + y^2)^{\frac{1}{2}}] r^2 \sqrt{x^2 + y^2} - f' r^2 (x^2 + y^2)^{\frac{1}{2}}}{r^4 (x^2 + y^2)}$$

$$= \frac{-2f' + f'' (x^2 + y^2)^{\frac{1}{2}}}{r^2 (x^2 + y^2)^{\frac{1}{2}}} = \frac{rf'' - 2f'}{r^3}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{(-f' + \frac{x f''}{r^3}) r^3 + x^2 f' (3r)}{r^6}$$

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{(-2f' + \frac{f''}{r}) r^3 + 3f' \cdot r^3}{r^6}$$

$$= \frac{r f'' + f''}{r^4}$$

1000. 12. 1. 32 $a=b=0$ 时 $f(x,y) = e^{-x} y^2$

32. 函数 $f(x,y) = e^{-x}(ax+b-y^2)$, 若 $f(-1,0)$ 为其极大值, 则 a, b 满足 $b=2a \geq 0$

$$f(x,y) = e^{-x}(ax+b-y^2)$$

$$\frac{\partial f}{\partial x} = -e^{-x}(ax+b-y^2) + e^{-x} \cdot a = e^{-x}(a-b-\frac{a}{-ax+y^2}) \stackrel{a-b=a}{\stackrel{b=2a}{\approx}} 0$$

$$\frac{\partial f}{\partial y} = -2ye^{-x} \stackrel{y=0}{=} 0 \Rightarrow y=0$$

$$A = \frac{\partial^2 f}{\partial x^2} = e^{-x}(b-a+ax-y^2-a) < 0 \quad b-3a < 0$$

$$B = \frac{\partial^2 f}{\partial x \partial y} = 2ye^{-x} = 0$$

$$C = \frac{\partial^2 f}{\partial y^2} = -2e^{-x} \stackrel{x=-1}{=} -2e$$

$$\Delta = AC-B^2 = -2e^2(b-2a-a) > 0 \quad b < 3a$$

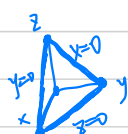
1000. 12. 1. 33

33. 曲面 $x^2 + 2y^2 + 3z^2 = 1$ 的切平面与三个坐标平面围成的有限区域的体积的最小值为

$$F(x,y,z) = x^2 + 2y^2 + 3z^2 - 1 = 0$$

$$\frac{\partial F}{\partial x} = 2x \quad \frac{\partial F}{\partial y} = 4y \quad \frac{\partial F}{\partial z} = 6z$$

$$\text{切平面 } G(X,Y,Z) = 2x(X-x) + 4y(Y-y) + 6z(Z-z) = 0$$



$$X=Y=0 \text{ 时 } Z = (2x^2 + 4y^2) / (6z + z) = \frac{2x^2 + 4y^2 + 6z^2}{6z}$$

$$Y=Z=0 \text{ 时 } X = (4y^2 + 6z^2) / (2x + x) = \frac{2x^2 + 4y^2 + 6z^2}{2x}$$

$$Z=X=0 \text{ 时 } Y = (6z^2 + 2x^2) / (4y + y) = \frac{2x^2 + 4y^2 + 6z^2}{4y}$$

$$V = \frac{1}{6} XYZ, \quad x^2 + 2y^2 + 3z^2 - 1 = 0$$

$$F(x,y,z,\mu) = \frac{1}{6} XYZ + \mu(x^2 + 2y^2 + 3z^2 - 1) = 0$$

$$\begin{cases} F'_x = \frac{(2x^2 + 4y^2 + 6z^2)^2}{3} \left[\frac{1}{24yz} \cdot \frac{x^2 - 2y^2 - 3z^2}{x^2} + \frac{1}{12xz} \cdot \frac{y}{y} + \frac{1}{8xy} \cdot \frac{2x}{3z} \right] + 2\mu x = 0 \\ F'_y = \frac{(2x^2 + 4y^2 + 6z^2)^2}{3} \left[\frac{4y}{24yz} \cdot \frac{x}{x} + \frac{1}{12xz} \cdot \frac{x^2 - 2y^2 - 3z^2}{2y^2} + \frac{1}{8xy} \cdot \frac{4y}{3z} \right] + 4\mu y = 0 \\ F'_z = \frac{(2x^2 + 4y^2 + 6z^2)^2}{3} \left[\frac{6z}{24yz} \cdot \frac{x}{x} + \frac{1}{12xz} \cdot \frac{3z}{y} + \frac{1}{8xy} \cdot \frac{x^2 - 2y^2 - 3z^2}{3z^2} \right] + 6\mu z = 0 \\ F'_\mu = x^2 + 2y^2 + 3z^2 - 1 = 0 \end{cases}$$

$$\begin{aligned} Z'_x &= \frac{4x}{6z} & Z'_y &= \frac{8y}{6z} & Z'_z &= \frac{-12x^2 - 24y^2 + 36z^2}{36z^2} \\ X'_x &= \frac{4x^2 - 8y^2 - 12z^2}{4x^2} & X'_y &= \frac{8y}{2x} & X'_z &= \frac{12z}{2x} \\ Y'_x &= \frac{4x}{4y} & Y'_y &= \frac{-8x^2 + 16y^2 - 24z^2}{16y^2} & Y'_z &= \frac{12z}{4y} \end{aligned}$$

$$\begin{aligned} Z'_x &= \frac{2x}{3z} & Z'_y &= \frac{4y}{3z} & Z'_z &= \frac{-x^2 - 2y^2 + 3z^2}{3z^2} \\ X'_x &= \frac{x^2 - 2y^2 - 3z^2}{x^2} & X'_y &= \frac{4y}{x} & X'_z &= \frac{6z}{x} \\ Y'_x &= \frac{x}{y} & Y'_y &= \frac{-x^2 + 2y^2 - 3z^2}{2y^2} & Y'_z &= \frac{3z}{y} \end{aligned}$$

$$\begin{cases} 2y^2 + 3z^2 = 1 - x^2 \\ x^2 + 3z^2 = 1 - 2y^2 \\ x^2 + 2y^2 = 1 - 3z^2 \end{cases}$$

$$\begin{cases} F'_x = \left[\frac{2}{3} - \frac{1}{12x^2} \right] + 3\mu xy z = 0 & 2\left(\frac{2}{3} - \frac{1}{12x^2}\right) = \frac{4}{3} - \frac{1}{12y^2} & 0 = \frac{1}{3x^2} - \frac{1}{6y^2} \Rightarrow x^2 = 2y^2 \\ F'_y = \left[\frac{4}{3} - \frac{1}{12y^2} \right] + 6\mu xy z = 0 & 3\left(\frac{2}{3} - \frac{1}{12x^2}\right) = \frac{2}{3} - \frac{1}{12z^2} & 0 = \frac{1}{2x^2} - \frac{1}{6z^2} \Rightarrow x^2 = 3z^2 \\ F'_z = \left[\frac{2}{3} - \frac{1}{12z^2} \right] + 9\mu xy z = 0 & & x^2 = 2y^2 = 3z^2 = \frac{1}{3} \end{cases}$$

$$V = \frac{1}{6} XYZ = \frac{1}{6} \cdot \frac{1}{x_0 \cdot 2y_0 \cdot 3z_0} = \frac{\sqrt{2}}{4}$$

$$\begin{aligned} 3z^2 &= 1 - x^2 - 2y^2 \\ 6z^2 &= 2 - 2x^2 - 4y^2 \end{aligned}$$

1000.13.1.34

34. 设函数 $f(x, y)$ 可微, $f'_y(x, y) = xf(x, y)$, $f(1, 0) = 1$, 且当 $x \neq 0$ 时, $\lim_{h \rightarrow 0} \left[\frac{f(x+h, 0)}{f(x, 0)} \right]^{\frac{1}{h}} = e^{\frac{1}{x}}$, 则 $f(x, y) =$ _____.

$$\begin{aligned} f'_y(1, 0) &= 1 \cdot f(1, 0) = 1 \\ \lim_{h \rightarrow 0} e^{\frac{1}{h} \frac{f(x+h, 0) - f(x, 0)}{f(x, 0)}} &= e^{\frac{f'_x(x, 0)}{f(x, 0)}} = e^{\frac{1}{x}} \\ f'_x(x, 0) &= \frac{f(x, 0)}{x} \rightarrow \frac{f'_x(x, 0)}{f(x, 0)} = \frac{1}{x} \rightarrow \ln |f(x, 0)| = \ln x + C \\ f'_x(1, 0) &= 1 \end{aligned}$$

$$\frac{f'_x(x, y)}{f(x, y)} = x \Rightarrow \ln |f(x, y)| = xy + \ln x$$

$$\begin{aligned} \ln |f(1, 0)| &= 0 = 0 + C \Rightarrow C = 0 \\ \Rightarrow \ln |f(x, 0)| &= \ln x \\ |f(x, 0)| &= x \end{aligned}$$

$$\varphi(x) = \ln x$$

$$\ln |f(x, y)| = xy + \ln x$$

$$|f(x, y)| = e^{xy + \ln x}$$

$$f(x, y) = \pm x e^{xy} \quad f(1, 0) = 1$$

$$f(x, y) = x e^{xy}$$

$$y^2 \left[\frac{1}{(x+1)^2} - \frac{1}{(1-x)^2} \right]$$

$$y^2 \left[\frac{1}{x+1} - \frac{1}{1-x} \right]$$

$$y^2 \left[(-1)(x+1)^{-2} - (1-x)^{-2} \right]$$

$$y^2 \left[(-1)(-2)(x+1)^{-3} - 2(1-x)^{-3} \right]$$

1000.13.1.35

35. 设 $z = y^2 \ln(1-x^2)$, 求 $\frac{\partial^n z}{\partial x^n} (n \geq 1)$.

$$z = y^2 \ln((1-x)(1+x)) = y^2 [\ln(1+x) + \ln(1-x)]$$

$$\frac{\partial^n z}{\partial x^n} = y^2 \left[(1+x)^{-n} (-1)^{n-1} - (1-x)^{-n} \right] (n-1)!$$

1000.13.1.36

36. 设 $z = e^{-x} - f(x-2y)$, 且当 $y = 0$ 时, $z = x^2$, 求 $\frac{\partial z}{\partial x}$.

$$x^2 = e^{-x} - \underline{f(x)} \Rightarrow f(x) = e^{-x} - x^2$$

$$z = e^{-x} - e^{-(x-2y)} + (x-2y)^2$$

$$\frac{\partial z}{\partial x} = -e^{-x} + e^{-(x-2y)} + 2(x-2y)$$

1000.13.1.37

37. 设函数 $u(x, y) = \varphi(x+y) + \varphi(x-y) + \int_{x-y}^0 \psi(t) dt$, 其中函数 φ 具有二阶导数, ψ 具有一阶导数, 求 $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2}$.

$$u'_x = \varphi'(x+y) + \varphi'(x-y) - \psi(x-y)$$

$$u'_y = \varphi'(x+y) - \varphi'(x-y) + \psi(x-y)$$

$$u''_{xx} = \varphi''(x+y) + \varphi''(x-y) - \psi'(x-y)$$

$$u''_{yy} = \varphi''(x+y) + \varphi''(x-y) - \psi'(x-y)$$

$$u''_{xx} - u''_{yy} = 0$$

1000.13.1.38

38. 设函数 $f(x, y, z) = e^x y z^2$, 其中 $z = z(x, y)$ 是由 $x + y + z + xyz = 0$ 确定的隐函数, 求 $f'_x(0, 1, -1)$.

$$f'_x = e^x y z^2 + e^x y \cdot 2z \cdot z'_x \Big|_{(0, 1, -1)} = 1$$

$$z'_x = - \frac{F'_x}{F'_z} = - \frac{1+y z}{1+xy} \Big|_{(0, 1, -1)} = 0$$

1000.13.1.39

39. 设 $u = f(x, y, z)$ 有连续的一阶偏导数, 且函数 $y = y(x)$ 及 $z = z(x)$ 分别由 $e^{xy} - xy = 2$ 和 $z = \int_0^x \sin t^2 dt$ 确定. 求 $\frac{du}{dx}$.

$$F(x, y) = e^{xy} - xy - 2 = 0$$

$$y'_x = -\frac{F'_x}{F'_y} = -\frac{ye^{xy} - y}{xe^{xy} - x} = -\frac{y}{x}$$

$$z'_x = \sin x^2$$

$$\frac{du}{dx} = f'_x - f'_y \frac{y}{x} + f'_z \cdot \sin x^2$$

1000.13.1.40

40. 设 $z = \sin(xy) + \varphi\left(x, \frac{x}{y}\right)$, 其中 φ 具有二阶连续偏导数, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

$$\frac{\partial z}{\partial x} = y \cos xy + \varphi'_1 + \frac{1}{y} \varphi'_2$$

$$\frac{\partial^2 z}{\partial x \partial y} = \cos xy - y \sin xy + \varphi''_{12} \left(-\frac{x}{y^2}\right) - \frac{x}{y^2} \cdot \frac{1}{y} \varphi''_{22} - \frac{1}{y^2} \varphi'_2$$

1000.13.1.41

41. 设 $u = yf\left(\frac{x}{y}\right) + xg\left(\frac{y}{x}\right)$, 其中函数 f, g 具有二阶连续导数, 求 $x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y}$.

$$\frac{\partial u}{\partial x} = y f' \cdot \frac{1}{y} + g + x g' \cdot \frac{-y}{x^2} = f' + g - \frac{y}{x} g'$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{f'}{y} + g' \left(-\frac{y}{x^2}\right) + \frac{y}{x^2} g' - \frac{y}{x} g'' \left(-\frac{y}{x^2}\right)$$

$$x \frac{\partial^2 u}{\partial x^2} = \frac{x}{y} f' + \frac{y}{x^2} g''$$

$$\frac{\partial^2 u}{\partial x \partial y} = f' \frac{-x}{y^2} + \frac{g'}{x} - \frac{g'}{x} - \frac{y}{x^2} g'' \quad + = 0$$

$$y \frac{\partial^2 u}{\partial x \partial y} = -\frac{x}{y} f' - \frac{y}{x^2} g''$$

1000.13.1.42

42. 设 $x = x(y, z)$, $y = y(z, x)$, $z = z(x, y)$ 均为由方程 $f(x, y, z) = 0$ 确定的具有连续偏导数的函数, 证明 $x'_y \cdot y'_z \cdot z'_x = -1$.

$$k'_y \cdot y'_z \cdot z'_x = \left(-\frac{f'_y}{f'_x}\right) \left(-\frac{f'_z}{f'_y}\right) \left(-\frac{f'_x}{f'_z}\right) = -1$$

1000.13.1.43

43. 设函数 $f(x, y)$ 可微, 又 $f(0, 0) = 0$, $f'_x(0, 0) = a$, $f'_y(0, 0) = b$, 且 $\varphi(t) = f[t, f(t, t^2)]$, 求 $\varphi'(0)$.

$$f(0, 0) = 0 \quad f'_x(0, 0) = a \quad f'_y(0, 0) = b$$

$$\varphi' = f'_1[t, f(t, t^2)] + f'_2[t, f(t, t^2)] \cdot [f'_1(t, t^2) + f'_2(t, t^2) \cdot 2t]$$

$$\begin{aligned} \varphi'(0) &= f'_x(0, 0) + f'_y(0, 0) \cdot [f'_x(0, 0)] \\ &= a + b \cdot a \end{aligned}$$

1000.13.1.44

44. 设 $u = f(x, y, z)$, $\varphi(x^2, e^x, z) = 0$, $y = \sin x$, 其中 f, φ 具有一阶连续偏导数, 且 $\frac{\partial \varphi}{\partial z} \neq 0$,

求 $\frac{du}{dx}$.

$$\left. \begin{aligned} \frac{du}{dx} &= f'_x + \frac{dy}{dx} f'_y + \frac{dz}{dx} f'_z \\ \frac{dy}{dx} &= \cos x \\ \frac{dz}{dx} &= -\frac{\varphi_x}{\varphi_z} = -\frac{\varphi'_1 \cdot 2x}{\varphi'_3} \end{aligned} \right\} \frac{du}{dx} = f'_x + \cos x f'_y - \frac{\varphi'_1}{\varphi'_3} \cdot 2x \cdot f'_z$$

1000.13.1.45

45. 已知 $\begin{cases} z = x^2 + y^2, \\ x^2 + 2y^2 + 3z^2 = 20, \end{cases}$ 求 $\frac{dy}{dx}, \frac{dz}{dx}$.

$$\begin{aligned} dz &= 2x dx + 2y dy & 2x dx + 4y dy + 6z dz &= 0 \\ \frac{dz}{dx} &= 2x + 2y \frac{dy}{dx} & 2x + 4y \frac{dy}{dx} + 6z \frac{dz}{dx} &= 0 \\ 2x + 4y \frac{dy}{dx} + 6z(2x + 2y \frac{dy}{dx}) &= 0 \\ 2x + 12xz + (4y + 12yz) \frac{dy}{dx} &= 0 \Rightarrow \frac{dy}{dx} = \frac{x(1+6z)}{y(2+6z)} \\ -2x + 6z \frac{dz}{dx} &= -2 \frac{dz}{dx} \Rightarrow \frac{dz}{dx} = \frac{x}{1+3z} \end{aligned}$$

1000.13.1.46

46. 设函数 $f(x, y)$ 具有二阶连续偏导数, 在点 $(1, 0)$ 的某邻域内, $f(x, y) = 1 - 2x + 3y + o(\sqrt{(x-1)^2 + y^2})$, 问: 函数 $g(x, y) = f(\cos x, x^2 + y^2)$ 在点 $(0, 0)$ 处是否取得极值? 若取得极值, 则判断取极大值还是极小值; 若不取得极值, 则说明理由.

$$\begin{aligned} f'_x &= -2 & f'_y &= 3 \\ g'_x &= 2\sin x + 6x & g'_x(0) &= 0 \text{ 是驻点} \\ g'_y &= 6y & g'_y(0) &= 0 \\ \begin{cases} g''_{xx} &= 2\cos x + 6 = A \\ g''_{xy} &= 0 = B \\ g''_{yy} &= 6 = C \end{cases} & \Delta = AC - B^2 = (2\cos x + 6) \cdot 6 \\ & \Delta(0, 0) = 48 > 0 & \text{极小值} \\ & A(0, 0) = 8 > 0 & \end{aligned}$$

1000.13.1.47

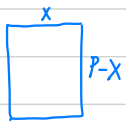
47. 设 $f(x, y)$ 在点 $O(0, 0)$ 处的某邻域 U 内连续, 且 $\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - xy}{x^2 + y^2} = a$, 常数 $a > \frac{1}{2}$. 讨论 $f(0, 0)$ 是否为 $f(x, y)$ 的极值, 若是极值, 判断是极大值还是极小值.

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - xy}{x^2 + y^2} = \frac{1}{x^2 + y^2} = \lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - 0}{x^2 + y^2} = f''(0, 0) = a > \frac{1}{2}$$

极小

1000.12.1.48

48. 已知矩形的周长为 $2p$, 将它绕其中一边旋转一周构成一个旋转体(圆柱体), 问该圆柱体的半径与高各为多少时, 该圆柱体体积最大?



$$\begin{aligned}
 V &= \pi x^2(p-x) \\
 V' &= 2x \cdot \pi p - 3\pi x^2 = \pi(2p-3x) \stackrel{?}{=} 0 \\
 \Rightarrow x &= \frac{2}{3}p \quad y = \frac{p}{3} \\
 V'' &= 2\pi p - 6\pi x \Big|_{\frac{2}{3}p} < 0
 \end{aligned}$$

1000.12.1.49

49. (1) 设 $x > 0, y > 0, z > 0$, 求函数 $f(x, y, z) = xyz^3$ 在约束条件 $x^2 + y^2 + z^2 = 5R^2 (R > 0$ 为常数) 下的最大值;

(2) 由(1)的结论证明: 当 $a > 0, b > 0, c > 0$ 时,

$$R^2 = \frac{x^2 + y^2 + z^2}{5}$$

$$abc^3 \leq 27 \left(\frac{a+b+c}{5} \right)^5$$

$$\begin{aligned}
 (1) \quad F(x, y, z, \lambda) &= xyz^3 + \lambda(x^2 + y^2 + z^2 - 5R^2) = 0 \\
 \begin{cases} F'_x = yz^3 + 2\lambda x = 0 \\ F'_y = xz^3 + 2\lambda y = 0 \\ F'_z = 3xy^2 + 2\lambda z = 0 \\ F'_\lambda = x^2 + y^2 + z^2 - 5R^2 = 0 \end{cases} & \Rightarrow \begin{cases} \lambda = \frac{-yz^3}{2x} \\ \lambda = \frac{-xz^3}{2y} \\ \lambda = \frac{-3xy^2}{2z} \end{cases} \\
 \frac{yz^3}{x} = \frac{xz^3}{y} = 3xy & \Rightarrow \begin{cases} x^2 = y^2 \\ z^2 = 3y^2 = 3x^2 \end{cases} \\
 5x^2 = 5R^2 \quad x = R \quad y = R \quad z = \sqrt{3}R & \\
 f(x, y, z) = 3\sqrt{3}R^5, \quad a=b=R \quad c=\sqrt{3}R &
 \end{aligned}$$

$$(2) \quad abc^3 \leq 3\sqrt{3}R^5$$

$$\begin{aligned}
 27 \left(\frac{a+b+c}{5} \right)^5 &\geq 27 \frac{3^5 (abc)^{\frac{5}{3}}}{5^5} \quad \frac{a+b+c}{3} \geq \sqrt[3]{abc} \\
 &= \left[27 \left(\frac{3}{5} \right)^5 3^{\frac{5}{6}} \right] R^5 \quad \frac{a+b}{2} \geq \sqrt{ab} \\
 &\approx 22
 \end{aligned}$$

$$\begin{aligned}
 xyz^3 &\leq 3\sqrt{3}R^5 = \left(\frac{x^2 + y^2 + z^2}{5} \right)^{\frac{5}{2}} \\
 \Rightarrow x^2 y^2 z^6 &\leq \left(\frac{x^2 + y^2 + z^2}{5} \right)^5 \\
 \Rightarrow a=x^2 \quad b=y^2 \quad c=z^2 \\
 \Rightarrow abc^3 &\leq \left(\frac{a+b+c}{5} \right)^5
 \end{aligned}$$

1000.12.1.50

50. 求 $f(x, y) = x + xy - x^2 - y^2$ 在闭区域 $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 2\}$ 上的最大值和最小值.

$$\begin{aligned}
 (1) \quad f(x, y) &= x + xy - x^2 - y^2 \\
 \begin{cases} f'_x = 1 + y - 2x \stackrel{?}{=} 0 \\ f'_y = x - 2y \stackrel{?}{=} 0 \\ f''_{xx} = -2 < 0 \\ f''_{yy} = -2 < 0 \\ f''_{xy} = 1 \quad D = 3 > 0 \end{cases} & \Rightarrow \begin{cases} y = 2x - 1 \\ x = 2y \end{cases} \Rightarrow \begin{cases} y = \frac{1}{3} \\ x = \frac{2}{3} \end{cases} \\
 \text{极大值} \quad f\left(\frac{2}{3}, \frac{1}{3}\right) &= \frac{2}{3} + \frac{2}{9} - \frac{4}{9} - \frac{1}{9} = \frac{1}{3} \quad \text{MAX}
 \end{aligned}$$



(2) $x=0$ 时

$$\begin{aligned}
 f(0, y) &= -y^2 \quad y \in [0, 2] \Rightarrow f_{\max 1} = f(0, 0) = 0 \\
 1 - 2y &= 0 \quad y = \frac{1}{2} \quad f_{\min 1} = f(0, \frac{1}{2}) = -\frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 x=1 \text{ 时 } f(1, y) &= y - y^2 \quad y \in [0, 2] \Rightarrow f_{\max 2} = f(1, 0) = 0 \\
 1 - 2y &= 0 \quad y = \frac{1}{2} \quad f_{\min 2} = f(1, \frac{1}{2}) = \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 y=0 \text{ 时 } f(x, 0) &= x - x^2 \quad x \in [0, 1] \Rightarrow f_{\max 3} = f(0, 0) = f(1, 0) = 0 \\
 3 - 2x &= 0 \quad x = \frac{3}{2} \quad f_{\min 3} = f(\frac{3}{2}, 0) = \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 y=2 \text{ 时 } f(x, 2) &= 3x - x^2 - 4 \quad x \in [0, 1] \Rightarrow f_{\max 4} = f(1, 2) = -2 \\
 3 - 2x &= 0 \quad x = \frac{3}{2} \quad f_{\min 4} = f(0, 2) = -4
 \end{aligned}$$

MIN

1000.13.1.51

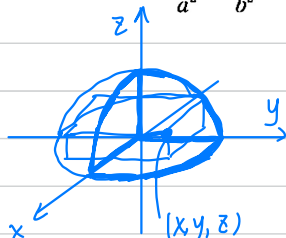
51. 求 $f(x, y) = 2x^2 + 3y^2$ 在有界闭区域 $D = \{(x, y) | (x-1)^2 + y^2 \leq 4\}$ 上的最大值与最小值.

① $f(x, y) = 2x^2 + 3y^2$
 $f'_x = 4x$
 $f'_y = 6y$
 $f''_{xx} = 4$
 $f''_{yy} = 6$
 $f''_{xy} = 0$
 $\Delta = AC - B^2 > 0$
 $A > 0$
 $f(0, 0) = 0$ 极小值
 $\min$

② $F(x, y, \lambda) = 2x^2 + 3y^2 + \lambda[(x-1)^2 + y^2 - 4]$
 $F'_x = 4x + 2\lambda(x-1) = 0$
 $F'_y = 6y + 2\lambda y = 0$
 $F'_\lambda = (x-1)^2 + y^2 - 4 = 0$
 $-8 = 6\lambda$
 $2x + \lambda x - \lambda = 0$
 $x = \frac{\lambda}{\lambda + 2}$
 $y = 0$
 $\lambda = -3$
 $x = 3$
 $y = 0$
 $f(3, 0) = 2 \times 9 = 18$
 $f(-1, 0) = 2 \times 1 = 2$
 $\max$

1000.13.1.52

52. 求内接于椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 (a, b, c > 0)$ 的长方体的最大体积.



$U = 8xyz$

$F(x, y, z, \lambda) = 8xyz + \lambda(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1) = 0$

$F'_x = 8yz + \frac{2\lambda x}{a^2} = 0$

$-8xyz = \frac{2\lambda x^2}{a^2} = \frac{2\lambda y^2}{b^2} = \frac{2\lambda z^2}{c^2}$

$x = \sqrt{\frac{a^2}{3}}$

$F'_y = 8xz + \frac{2\lambda y}{b^2} = 0$

$\Rightarrow \lambda = 0$ 时 $\lambda \neq 0$ 时 $\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} = \frac{1}{3}$

$F'_z = 8xy + \frac{2\lambda z}{c^2} = 0$

$F'_\lambda = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$

$U = 8xyz = 8 \cdot \frac{abc}{3\sqrt{3}}$

1000.13.1.53

53. 在第一象限的椭圆 $\frac{x^2}{4} + y^2 = 1$ 上求一点, 使过该点的法线与原点的距离最大.



$F(x, y) = \frac{x^2}{4} + y^2 - 1 = 0$

$\vec{r} = (\frac{x}{2}, 2y)$

$\vec{n} = (-2y, \frac{x}{2})$

$-2y(x - \frac{x}{2}) + \frac{x}{2}(2y - y) = 0$

$d = \frac{|2yx - \frac{xy}{2}|}{\sqrt{4y^2 + \frac{x^2}{4}}} = \frac{3|xy|}{\sqrt{16y^2 + x^2}}$

$12 - 48$

$(12y^2 - 3x^2)(16y^2 + x^2) = -36x^2y^2$

$12 \times 16 y^4 = 3x^4$

$64y^4 = x^4 \Rightarrow \sqrt{2}y = x$

$\begin{cases} 12y^2\sqrt{16y^2 + x^2} - 12x^2y^2(16y^2 + x^2)^{-\frac{1}{2}} = 2\lambda xy(16y^2 + x^2) \\ 3x\sqrt{16y^2 + x^2} - 48xy^2(16y^2 + x^2)^{-\frac{1}{2}} = 2\lambda xy(16y^2 + x^2) \end{cases}$

$d = \frac{\frac{3\sqrt{2}}{4}}{\frac{2\sqrt{2}}{2\sqrt{2}}} = \frac{3}{4}$

1000.13.1.54

54. 求正数 a, b 的值, 使得椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 包含圆 $x^2 + y^2 = 2y$, 且面积最小.

54. 【解析】如图 1-13-1 所示, 所求椭圆必须包含圆 $x^2 + y^2 = 2y$, 并与之相切. 故在椭圆上的任意一点 (x, y) 处满足 $f(x, y) = x^2 + (y-1)^2 \geq 1$. 这说明函数 $f(x, y) = x^2 + (y-1)^2$ 在椭圆方程 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 的约束下取得最小值为 1. 于是考虑条件极值问题:

$\begin{cases} \min\{f(x, y)\} = 1, \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \end{cases}$

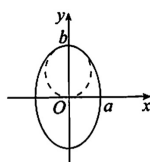


图 1-13-1

$F(x, y, \lambda) = x^2 + (y-1)^2 + \lambda(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)$

$F'_x = 2x + \frac{2\lambda}{a^2}x = 0$ $x \neq 0$ 时 $\lambda = -a^2$

$F'_y = 2(y-1) + \frac{2\lambda}{b^2}y = 0$ $\lambda = -a^2$

$F'_\lambda = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$ $y^2 = \frac{b^4}{b^2 - a^2}$ $y = \frac{b^2}{b^2 - a^2}$ $x^2 = (1 - \frac{b^2}{b^2 - a^2})a^2$ $y-1 = \frac{b^2 - a^2}{b^2 - a^2}$

$x^2 + (y-1)^2 = (1 - \frac{b^2}{b^2 - a^2})a^2 + (\frac{b^2 - a^2}{b^2 - a^2})^2 = 1$

$$\frac{[(b^2-a^2)^2 - b^2] a^2 + a^4}{(b^2-a^2)^2} = 1$$

$$\pi ab = \pi [\sqrt{6}]$$

(000.13.1.55)

55. 求椭球体 $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{4} \leq 1$ 被平面 $x+y+z=0$ 所截得的椭圆的面积.

$$d = \sqrt{2} \quad d = \sqrt{2}$$

$$\Rightarrow (1, 1, 0) \quad (-1, 1, 0)$$

$$x^2 + y^2 = 2$$

$$d = \sqrt{x^2 + y^2 + z^2} \quad \min, \max$$

$$[b^4 + a^4 - 2a^2b^2 - b^2] a^2 + a^4 = (b^2 - a^2)^2$$

$$a^2b^4 + a^6 - 2a^4b^2 - a^2b^2 + a^4 = a^4 + b^4 - 2a^2b^2$$

$$a^2b^4 + a^6 - 2a^4b^2 = b^4 - a^2b^2$$

$$a^2[b^4 + a^4 - 2a^2b^2] = a^2(a^2 - b^2)^2 = b^2(b^2 - a^2)$$

$$a^2b^4 - a^4b^2 = 0$$

$$\Rightarrow H(a, b, \eta) = ab + \eta(a^2b^2 - a^4 - b^4)$$

$$H'_a = b + \eta(2ab^2 - 4a^3) = b + a\eta(2b^2 - 4a^2)$$

$$H'_b = a + \eta(2a^2b - 2b^3) = a + b\eta(2a^2 - 2b^2)$$

$$H'_\eta = a^2b^2 - a^4 - b^4 = 0$$

$$a^2 \cdot 2a^2 - 3a^4 = 0 \quad \Rightarrow \quad a^2 = 3$$

$$b^2 = 2a^2 = 6 \quad \Rightarrow \quad b = \sqrt{6}$$

(000.13.1.56)

56. 已知函数 $u = u(x, y)$ 满足方程 $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + k\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\right) = 0$. 确定参数 a, b , 利用变换

$u(x, y) = v(x, y)e^{ax+by}$ 将原方程变形, 使新方程中不含有一阶偏导数项.

$$u'_x = v'_x e^{ax+by} + v \cdot e^{ax+by} \cdot a$$

$$u'_y = v'_y e^{ax+by} + v \cdot e^{ax+by} \cdot b$$

$$u''_{xx} = v''_{xx} e^{ax+by} + 2v'_x e^{ax+by} \cdot a + v \cdot e^{ax+by} \cdot a^2$$

$$u''_{yy} = v''_{yy} e^{ax+by} + 2v'_y e^{ax+by} \cdot b + v \cdot e^{ax+by} \cdot b^2$$

$$(v''_{xx} - v''_{yy}) e^{ax+by} + 2e^{ax+by} (av'_x - bv'_y) + v \cdot e^{ax+by} (a^2 - b^2) + ke^{ax+by} (v'_x + v'_y) + v \cdot e^{ax+by} (a + b) = 0$$

$$\Rightarrow 2(av'_x - bv'_y) + k(v'_x + v'_y) = 0$$

$$2a + k = 0 \quad -2b + k = 0$$

(000.13.1.57)

57. 设 A, B, C 为常数, $AC - B^2 < 0, A \neq 0, u(x, y)$ 具有二阶连续偏导数. 证明: 必存在非奇异线性变换

$$\xi = \lambda_1 x + y, \eta = \lambda_2 x + y (\lambda_1, \lambda_2 \text{ 为常数}),$$

将方程 $A \frac{\partial^2 u}{\partial x^2} + 2B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} = 0$ 化成 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$.

$$\xi = \lambda_1 x + y$$

$$\eta = \lambda_2 x + y$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x}$$

$$\Rightarrow \begin{cases} u'_x = \lambda_1 u'_\xi + \lambda_2 u'_\eta \\ u'_y = u'_\xi + u'_\eta \end{cases}$$

$$A \begin{cases} u''_{xx} = \lambda_1^2 u''_{\xi\xi} + \lambda_2^2 u''_{\eta\eta} + 2\lambda_1 \lambda_2 u''_{\xi\eta} \\ u''_{xy} = \lambda_1 u''_{\xi\xi} + \lambda_2 u''_{\xi\eta} + (\lambda_1 + \lambda_2) u''_{\eta\xi} \\ u''_{yy} = u''_{\xi\xi} + u''_{\eta\eta} + 2u''_{\xi\eta} \end{cases}$$

$$B \begin{cases} u''_{xy} = \lambda_1 u''_{\xi\xi} + \lambda_2 u''_{\xi\eta} + (\lambda_1 + \lambda_2) u''_{\eta\xi} \\ u''_{yy} = u''_{\xi\xi} + u''_{\eta\eta} + 2u''_{\xi\eta} \end{cases}$$

$$C \begin{cases} u''_{xy} = \lambda_1 u''_{\xi\xi} + \lambda_2 u''_{\xi\eta} + (\lambda_1 + \lambda_2) u''_{\eta\xi} \\ u''_{yy} = u''_{\xi\xi} + u''_{\eta\eta} + 2u''_{\xi\eta} \end{cases}$$

$$A u''_{xx} + 2B u''_{xy} + C u''_{yy} = (A\lambda_1^2 + 2B\lambda_1 + C) u''_{\xi\xi} + (A\lambda_2^2 + 2B\lambda_2 + C) u''_{\eta\eta} + (2A\lambda_1\lambda_2 + 2B(\lambda_1 + \lambda_2) + C) u''_{\xi\eta}$$

$$\Delta = 4B^2 - 4AC = 4(B^2 - AC) > 0$$

$$\text{存在 } \lambda_1, \lambda_2 \text{ 使 } A\lambda^2 + 2B\lambda + C = 0$$

$$\Rightarrow A u''_{xx} + 2B u''_{xy} + C u''_{yy} = (2A\lambda_1 + 2B\lambda_2 + 2A\lambda_1\lambda_2 + 2C) u''_{\xi\eta} = 0 \Rightarrow u''_{\xi\eta} = 0$$

1000.17.1.58

58. 设 $z = z(u, v)$ 具有二阶连续偏导数, 且 $z = z(x+y, x-y)$ 满足微分方程

$$\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 1.$$

(1) 求 $z = z(u, v)$ 所满足关于 u, v 的微分方程;

(2) 由(1) 求出 $z = z(x+y, x-y)$ 的一般表达式.

$$z'_x = z'_1 + z'_2$$

$$z'_y = z'_1 - z'_2$$

$$z''_{xx} = z''_{11} + 2z''_{12} + z''_{22}$$

$$z''_{yy} = z''_{11} - 2z''_{12} + z''_{22}$$

$$2 \quad z''_{xy} = z''_{11} - z''_{22}$$

$$z''_{xx} + 2z''_{xy} + z''_{yy} = 4z''_{11} = 1$$

$$\Rightarrow z''_{uu} = \frac{1}{4}$$

$$(2) \quad z'_u = \frac{u}{4} + \varphi(v)$$

$$z = \frac{u^2}{8} + \varphi(v) \cdot u + \psi(v)$$

$$z = \frac{(x+y)^2}{8} + \varphi(x-y) \cdot (x+y) + \psi(x-y)$$